

DIFFERENTIAL EQUATIONS

From Local Change to Evolution

Calculus, Dynamics, and Waves

Extended mathematical problem

Complete ten-part paper

Extended paper — intended for several sittings

Calculators and electronic aids are not required.

Adapted from the *Differential Equations* notes based on lectures by M. G. Worster. Original notes by Dexter Chua.

Instructions and planned structure

This completed paper consists of one problem divided into the following ten parts.

- I. Differentiation, asymptotics, and local approximation;
- II. integration and multivariable differentiation;
- III. first-order linear differential equations;
- IV. nonlinear first-order equations and discrete dynamics;
- V. homogeneous second-order equations and phase space;
- VI. forced and equidimensional equations, recurrences, damping, and impulses;
- VII. series solutions near ordinary and singular points;
- VIII. gradients, Taylor forms, and stationary points;
- IX. differential systems and phase portraits;
- X. characteristics, waves, and diffusion.

Question numbers remain continuous across all ten parts. The result of any question may be used later, even if that question has not been answered, provided its number is cited clearly. Indications are optional. Definitions and notation are introduced when they first become necessary. No result may be used unless it belongs to the calculus background specified below, has been stated as an admitted result, or has been proved in an earlier question.

Part I. — Differentiation, asymptotics, and local approximation

As in the original notes, the reader is assumed to know elementary single-variable real calculus: real limits and their algebra, continuity, the usual elementary functions and their derivatives, the mean-value theorem, and mathematical induction. The definitions and results emphasized by the course are nevertheless stated and developed here.

A. — Derivatives and equations of evolution

Derivative and differentiability. Let $I \subseteq \mathbb{R}$ be an open interval, let $f: I \rightarrow \mathbb{R}$, and let $x_0 \in I$. The *derivative* of f at x_0 is

$$f'(x_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + h) - f(x_0)}{h},$$

provided this real limit exists. In that case, f is *differentiable at x_0* . It is differentiable on I if it is differentiable at every $x_0 \in I$. The right-hand and left-hand derivatives are defined by replacing $h \rightarrow 0$ with $h \rightarrow 0^+$ and $h \rightarrow 0^-$, respectively. Thus f is differentiable at x_0 precisely when both one-sided derivatives exist and are equal.

1)

- a) **DQPL** *Difference Quotient for a Power Law.* Define $p: \mathbb{R} \rightarrow \mathbb{R}$ by $p(x) = x^2$ for every $x \in \mathbb{R}$. Starting from the difference quotient, prove that $p'(x_0) = 2x_0$ for every $x_0 \in \mathbb{R}$.
- b) **ONDD** *One-sided Non-Differentiability Demonstration.* Define $a: \mathbb{R} \rightarrow \mathbb{R}$ by $a(x) = |x|$ for every $x \in \mathbb{R}$. Compute the right-hand and left-hand derivatives of a at 0 and prove that a is not differentiable there.
- c) **HDC** *Higher Derivative Conventions.* When f is differentiable, its derivative function is also written

$$\frac{df}{dx} = f'(x) = \frac{d}{dx} f(x).$$

If f' is differentiable, define $f'' = (f')'$ and write $d^2f/dx^2 = f''(x)$. Let $f: I \rightarrow \mathbb{R}$ be differentiable, put $J = \{t \in \mathbb{R} : 2t \in I\}$, and define $g: J \rightarrow \mathbb{R}$ by $g(t) = f(2t)$. Use the difference quotient to prove $g'(t) = 2f'(2t)$ for every $t \in J$.

Differential and difference equations. Let $I \subseteq \mathbb{R}$ be an interval. A *differential equation* is an equation imposed on an unknown function $x: I \rightarrow \mathbb{R}$ and some of its derivatives. A *solution on I* is a function possessing the required derivatives and satisfying the equation at every $t \in I$. When the argument is time t , write

$$\dot{x}(t) = \frac{dx}{dt}(t), \quad \ddot{x}(t) = \frac{d^2x}{dt^2}(t).$$

An *initial condition* prescribes one or more values of the unknown function and its derivatives at a specified time.

For equations involving an unknown twice differentiable function $x: \mathbb{R} \rightarrow \mathbb{R}$, call an equation *linear* if it can be written

$$a_2(t)\ddot{x}(t) + a_1(t)\dot{x}(t) + a_0(t)x(t) = b(t) \quad (t \in \mathbb{R}),$$

where $a_0, a_1, a_2, b: \mathbb{R} \rightarrow \mathbb{R}$ are prescribed functions independent of x . An equation that cannot be written in this form is *nonlinear*.

2) Let $m \in \mathbb{R}$ satisfy $m > 0$. In a one-dimensional Newtonian model, $t \in \mathbb{R}$ denotes time, a twice differentiable function $x: \mathbb{R} \rightarrow \mathbb{R}$ gives the particle's position, and its force is the function $F_x: \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$F_x(t) = m\ddot{x}(t) \quad (t \in \mathbb{R}).$$

- a) **PADE** *Prescribed Acceleration Differential Equation.* Suppose that $F_x(t) = t^2 - 1$ for every $t \in \mathbb{R}$. Find all possible position functions, or equivalently all twice differentiable functions $x: \mathbb{R} \rightarrow \mathbb{R}$ satisfying

$$m\ddot{x}(t) = t^2 - 1 \quad (t \in \mathbb{R}).$$

Prove that they are exactly the functions

$$x(t) = \frac{t^4}{12m} - \frac{t^2}{2m} + c_1t + c_0 \quad (t \in \mathbb{R}),$$

where $c_0, c_1 \in \mathbb{R}$.

- b) **PDNDE** *Position-Dependent Nonlinear Differential Equation.* Suppose instead that $F_x(t) = x(t)^2 - 1$ for every $t \in \mathbb{R}$, so that

$$m\ddot{x}(t) = x(t)^2 - 1 \quad (t \in \mathbb{R}).$$

Explain why the occurrence of $x(t)^2$ makes this equation nonlinear. Determine all constant solutions $x: \mathbb{R} \rightarrow \mathbb{R}$.

- c) **ICSU** *Initial Conditions Select Uniquely.* Let $x_0, v_0 \in \mathbb{R}$. Among the solutions in part (a), find the unique one satisfying $x(0) = x_0$ and $\dot{x}(0) = v_0$.

Indication.— For the converse statement in part (a), subtract the displayed candidate from an arbitrary solution and use the standard consequence of the mean-value theorem that a function with zero second derivative is affine.

Write $\mathbb{N} = \{1, 2, 3, \dots\}$ and $\mathbb{N}_0 = \{0, 1, 2, \dots\}$. A real sequence is a map $F: \mathbb{N}_0 \rightarrow \mathbb{R}$, whose value at $n \in \mathbb{N}_0$ is denoted by F_n . A *difference equation* is a relation between terms of an unknown sequence. A solution is a sequence satisfying that relation for every index for which it is stated.

- 3) Consider the Fibonacci difference equation

$$F_{n+2} - F_{n+1} - F_n = 0 \quad (n \in \mathbb{N}_0).$$

- a) **FRCS** *Fibonacci Recurrence Computes Successors.* For the solution with $F_0 = 0$ and $F_1 = 1$, compute $F_2, F_3, \dots, F_7$.
- b) **TRIVFS** *Two Recurrence Initial Values Fix the Sequence.* Let $F, G: \mathbb{N}_0 \rightarrow \mathbb{R}$ be two solutions. Prove by induction that $F_0 = G_0$ and $F_1 = G_1$ imply $F_n = G_n$ for every $n \in \mathbb{N}_0$.
- c) **RANC** *Recurrence Alone is Not Complete.* Exhibit two distinct solutions of the difference equation. Explain why the recurrence without two initial values does not specify a unique sequence.

The explicit formula for the Fibonacci solution will be derived in Part VI. These first examples also illustrate a recurring method: one may guess a candidate solution, but the candidate becomes a solution only after its domain, differentiability, and governing equation have been checked.

B. — Small and bounded asymptotic orders

Asymptotic notation. Let $I \subseteq \mathbb{R}$ be an interval, let $x_0 \in \mathbb{R} \cup \{-\infty, +\infty\}$ be a limit point of I in the relevant direction, and let $f, g: I \rightarrow \mathbb{R}$. All limits below are taken through points $x \in I$. A *deleted neighbourhood* of a finite x_0 consists of the points satisfying $0 < |x - x_0| < \delta$ for some $\delta > 0$; at $+\infty$ it consists of the points satisfying $x > M$ for some $M \in \mathbb{R}$, and at $-\infty$ it is defined analogously.

Suppose first that $g(x) \neq 0$ throughout some deleted neighbourhood of x_0 . We write

$$f(x) = o(g(x)) \quad \text{as } x \rightarrow x_0$$

if

$$\lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} = 0.$$

We write

$$f(x) = O(g(x)) \quad \text{as } x \rightarrow x_0$$

if there exist $C \in \mathbb{R}$ with $C \geq 0$ and a deleted neighbourhood of x_0 throughout which

$$|f(x)| \leq C|g(x)|.$$

The latter definition remains meaningful without the non-vanishing assumption on g . Informally, $o(g)$ is negligible compared with g , whereas $O(g)$ is bounded in magnitude by a constant multiple of g .

4)

- a) **SROZ** *Square-Root Order near Zero*. On the interval $(0, \infty)$, prove that $x = o(\sqrt{x})$ as $x \rightarrow 0^+$.
- b) **SRGI** *Square Root Growth at Infinity*. On the interval $(0, \infty)$, prove that $\sqrt{x} = o(x)$ as $x \rightarrow +\infty$.
- c) **LSOB** *Local Sine Order Bound*. Using the elementary inequality $|\sin u| \leq |u|$ for every $u \in \mathbb{R}$, prove that $\sin(2x) = O(x)$ as $x \rightarrow 0$.
- d) **SBI** *Sine Bounded at Infinity*. Prove that $\sin(2x) = O(1)$ as $x \rightarrow +\infty$, although $\sin(2x)$ has no limit as $x \rightarrow +\infty$.
- e) **VRGBR** *Vanishing Ratio Gives a Bounded Ratio*. Let $f, g: I \rightarrow \mathbb{R}$ satisfy the hypotheses in the definition of small order. Prove that

$$f(x) = o(g(x)) \implies f(x) = O(g(x)) \quad (x \rightarrow x_0).$$

- f) **ONCRF** *Order Notation denotes a Class Rather than a Function*. Explain why $o(g)$ denotes a class of functions rather than one function. Interpret the conventional equality $f = o(g)$ as a membership assertion, and state that assertion using the symbol $\in$.

5) In every part, all functions are real-valued and all asymptotic statements refer to the same limit $x \rightarrow x_0$.

- a) **SRAC** *Scaling Rules for Asymptotic Classes*. Let $\lambda \in \mathbb{R}$. Prove that $f = o(g)$ implies $\lambda f = o(g)$, and that $f = O(g)$ implies $\lambda f = O(g)$.
- b) **SAR** *Sum of Asymptotic Remainders*. Prove that $f_1 = o(g)$ and $f_2 = o(g)$ imply $f_1 + f_2 = o(g)$. Prove the analogous assertion with O in place of o .
- c) **PROB** *Product Remainder Order Bound*. Prove that $f = O(g)$ and $p = O(q)$ imply $fp = O(gq)$. Deduce that, as $h \rightarrow 0$, the relations $r(h) = o(h)$ and $s(h) = O(h)$ imply $r(h)s(h) = o(h^2)$.
- d) **CRCL** *Composition Remainder Control Lemma*. Let $\delta > 0$, let $k: (-\delta, \delta) \rightarrow \mathbb{R}$ satisfy $k(h) \rightarrow 0$ and $k(h) = O(h)$ as $h \rightarrow 0$, and let ε be a real-valued function defined on an open interval containing 0, with $\varepsilon(u) \rightarrow 0$ as $u \rightarrow 0$. Prove

$$k(h)\varepsilon(k(h)) = o(h) \quad (h \rightarrow 0).$$

6) Let $I \subseteq \mathbb{R}$ be an open interval, let $x_0 \in I$, and let $f: I \rightarrow \mathbb{R}$.

- a) **DILA** *Derivative Implies Linear Approximation*. Suppose that f is differentiable at x_0 . Starting from the derivative limit, prove

$$f(x_0 + h) = f(x_0) + f'(x_0)h + o(h) \quad (h \rightarrow 0).$$

- b) **LADD** *Linear Approximation Determines the Derivative.* Conversely, suppose that $A \in \mathbb{R}$ and a function r , defined near 0, satisfy

$$f(x_0 + h) = f(x_0) + Ah + r(h), \quad r(h) = o(h) \quad (h \rightarrow 0).$$

Prove that f is differentiable at x_0 and $f'(x_0) = A$.

- c) **LAC** *Linear Approximation Coefficient.* Prove that at most one number $A \in \mathbb{R}$ can occur as the coefficient of h in an expansion of the form in part (b).
d) **QEO** *Quadratic Error Order.* For $f(x) = x^2$ on $\mathbb{R}$, show directly that

$$f(x_0 + h) = f(x_0) + 2x_0h + h^2.$$

Prove that $h^2 = O(h^2)$ and $h^2 = o(h)$ as $h \rightarrow 0$.

C. — Product, composition, and repeated differentiation

- 7) Let $I \subseteq \mathbb{R}$ be an open interval, let $x_0 \in I$, and let $u, v: I \rightarrow \mathbb{R}$ be differentiable at x_0 .

- a) **PRE** *Product Remainder Expansion.* Introduce functions r_u, r_v , defined near 0, such that $r_u(h) = o(h)$, $r_v(h) = o(h)$, and

$$u(x_0 + h) = u(x_0) + hu'(x_0) + r_u(h),$$

$$v(x_0 + h) = v(x_0) + hv'(x_0) + r_v(h).$$

Multiply these two identities.

- b) **CTRC** *Cross-Term Remainder Control.* Using Question 5, prove that all terms in the product from part (a), other than the constant term and the terms linear in h , have sum $o(h)$.
c) **DPR** *Derivative Product Rule.* Apply Question 6 to prove that

$$(uv)'(x_0) = u'(x_0)v(x_0) + u(x_0)v'(x_0).$$

- 8) Let $I, J \subseteq \mathbb{R}$ be open intervals, let $g: I \rightarrow J$, let $F: J \rightarrow \mathbb{R}$, and let $x_0 \in I$. Suppose that g is differentiable at x_0 and F is differentiable at $y_0 = g(x_0)$.

- a) **ICI** *Inner Composition Increment.* For all sufficiently small $h \in \mathbb{R}$, put $k(h) = g(x_0 + h) - g(x_0)$. Prove that

$$k(h) = hg'(x_0) + o(h), \quad k(h) = O(h), \quad k(h) \rightarrow 0 \quad (h \rightarrow 0).$$

- b) **ORF** *Outer Remainder Function.* For every sufficiently small $s \in \mathbb{R}$ with $y_0 + s \in J$, define

$$\varepsilon(s) = \begin{cases} \frac{F(y_0 + s) - F(y_0) - F'(y_0)s}{s}, & s \neq 0, \\ 0, & s = 0. \end{cases}$$

Prove that $\varepsilon(s) \rightarrow 0$ as $s \rightarrow 0$.

- c) **CRB** *Composed Remainder Bound.* Prove

$$F(g(x_0 + h)) - F(g(x_0)) = F'(y_0)k(h) + k(h)\varepsilon(k(h)).$$

Use Question 5(d) to obtain a linear approximation in h .

- d) **DCR** *Derivative Chain Rule.* Conclude that $F \circ g$ is differentiable at x_0 and

$$(F \circ g)'(x_0) = F'(g(x_0))g'(x_0).$$

Repeated derivatives and binomial coefficients. Let $I \subseteq \mathbb{R}$ be an open interval and let $f: I \rightarrow \mathbb{R}$. For $n \in \mathbb{N}_0$, set $f^{(0)} = f$ and, whenever the required derivatives exist, define $f^{(n+1)} = (f^{(n)})'$. Define $0! = 1$ and $n! = 1 \cdot 2 \cdots n$ for $n \in \mathbb{N}$. If $n \in \mathbb{N}_0$ and $r \in \mathbb{N}_0$ satisfies $0 \leq r \leq n$, define

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}.$$

9) Let $I \subseteq \mathbb{R}$ be an open interval and let $u, v: I \rightarrow \mathbb{R}$ possess every derivative that occurs below.

a) **ZOPC** *Zeroth and Onefold Product Cases.* Verify that

$$(uv)^{(n)} = \sum_{r=0}^n \binom{n}{r} u^{(r)} v^{(n-r)} \quad (L_n)$$

holds for $n = 0$ and $n = 1$.

b) **DIH** *Differentiated Induction Hypothesis.* Let $n \in \mathbb{N}$. Assume (L_n) and differentiate its two sides. Use the product rule to obtain two finite sums representing $(uv)^{(n+1)}$.

c) **PCR** *Pascal Combination after Reindexing.* Reindex one of the sums from part (b), and prove directly from the factorial definition that

$$\binom{n}{r} + \binom{n}{r-1} = \binom{n+1}{r} \quad (1 \leq r \leq n).$$

Combine the two sums, treating the terms with $r = 0$ and $r = n + 1$ separately.

d) **GLDFP** *General Leibniz Derivative Formula for Products.* Use induction to prove that, for every $n \in \mathbb{N}_0$ and all n -times differentiable functions $u, v: I \rightarrow \mathbb{R}$,

$$(uv)^{(n)}(x) = \sum_{r=0}^n \binom{n}{r} u^{(r)}(x) v^{(n-r)}(x) \quad (x \in I).$$

Indication.— In part (c), replace $r+1$ by a new index in the sum containing $u^{(r+1)}v^{(n-r)}$ before combining coefficients.

D. — Taylor polynomials and formal series

Taylor polynomial. Let $n \in \mathbb{N}_0$, let $I \subseteq \mathbb{R}$ be open, let $x_0 \in I$, and let $f: I \rightarrow \mathbb{R}$ possess derivatives through order n at x_0 . The *Taylor polynomial of order n about x_0* is

$$T_{n,x_0}f(x) = \sum_{r=0}^n \frac{f^{(r)}(x_0)}{r!} (x - x_0)^r \quad (x \in \mathbb{R}).$$

Admitted result. **TPPR** *Taylor Theorem with Peano Remainder.* Let $n \in \mathbb{N}_0$, let $I \subseteq \mathbb{R}$ be an open interval, let $x_0 \in I$, and let $f: I \rightarrow \mathbb{R}$. Suppose that the derivatives $f^{(0)}, f^{(1)}, \dots, f^{(n)}$ exist on a neighbourhood of x_0 and that $f^{(n)}$ is continuous at x_0 . Then, as $h \rightarrow 0$ with $x_0 + h \in I$,

$$f(x_0 + h) = \sum_{r=0}^n \frac{f^{(r)}(x_0)}{r!} h^r + o(h^n).$$

If, in addition, $f^{(n+1)}$ exists and is bounded on a neighbourhood of x_0 , the remainder after the displayed finite sum is $O(h^{n+1})$.

10) Retain the notation and hypotheses of the admitted Taylor theorem.

a) **FTPC** *Finite Taylor Polynomial Construction.* Prove that, for every integer $k \in \mathbb{N}_0$ with $0 \leq k \leq n$,

$$(T_{n,x_0}f)^{(k)}(x_0) = f^{(k)}(x_0).$$

b) **EVTF** *Equivalent Variable Taylor Form.* Put $x = x_0 + h$ and rewrite the admitted result as an expansion of $f(x)$ as $x \rightarrow x_0$. State the remainder in both the o form and, under the additional hypothesis, the O form.

- c) **LNAL** *Local Nature and Analytic Limitation*. Explain precisely why a statement as $x \rightarrow x_0$ controls f only near x_0 . Explain also why the admitted theorem, for one fixed finite $n \in \mathbb{N}_0$, does not assert that an infinite series converges to $f(x)$.
- d) **CTE** *Cubic Taylor Expansion*. Define $f: \mathbb{R} \rightarrow \mathbb{R}$ by $f(x) = x^3$. Compute $T_{3,1}f(x)$ and verify directly that it equals $f(x)$ for every $x \in \mathbb{R}$.

Taylor series. Let $I \subseteq \mathbb{R}$ be open, let $x_0 \in I$, and let $f: I \rightarrow \mathbb{R}$ possess derivatives of every order at x_0 . The *Taylor series of f about x_0* is the formal power series

$$\sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!} (x - x_0)^n.$$

The Taylor series about $x_0 = 0$ is also called the *Maclaurin series*. The word *formal* means that this definition alone asserts neither convergence of the series nor equality of its sum with $f(x)$.

11)

- a) **PTSF** *Polynomial Taylor Series is Finite*. Let $m \in \mathbb{N}_0$, let $a_0, \dots, a_m \in \mathbb{R}$, and define $p: \mathbb{R} \rightarrow \mathbb{R}$ by

$$p(x) = \sum_{r=0}^m a_r x^r.$$

Prove that the Taylor series of p about 0 has no non-zero terms after degree m and is equal to $p(x)$ for every $x \in \mathbb{R}$.

- b) **FSD** *Formal Series Distinction*. State separately the following three assertions about a Taylor series: its coefficients are defined; the resulting numerical series converges at a specified $x \in I$; its sum at that x equals $f(x)$. Explain which assertion follows from the definition alone.
- c) **MTRF** *Maclaurin Terms for a Rational Function*. Define $f: \mathbb{R} \setminus \{1\} \rightarrow \mathbb{R}$ by $f(x) = (1 - x)^{-1}$. Prove that $f^{(n)}(0) = n!$ for every $n \in \mathbb{N}_0$ and hence determine its formal Maclaurin series.

E. — Quotients with vanishing numerator and denominator

An expression $f(x)/g(x)$ has the *indeterminate form* $0/0$ at $x_0 \in \mathbb{R}$ when $f(x) \rightarrow 0$ and $g(x) \rightarrow 0$ as $x \rightarrow x_0$. The phrase does not assert that the quotient has a limit.

12) Let $I \subseteq \mathbb{R}$ be an open interval, let $x_0 \in I$, and let $f, g: I \rightarrow \mathbb{R}$ be differentiable at x_0 . Suppose $f(x_0) = g(x_0) = 0$ and $g'(x_0) \neq 0$.

- a) **FLQE** *First-order Linear Quotient Expansions*. Use Question 6 to write expansions of $f(x_0 + h)$ and $g(x_0 + h)$ to first order, with remainders in $o(h)$.
- b) **PNNZD** *Punctured-Neighbourhood Non-Zero Denominator*. Prove that there exists $\delta > 0$ such that $g(x_0 + h) \neq 0$ whenever $0 < |h| < \delta$.
- c) **LQDL** *Local Quotient Derivative Limit*. Divide the expansions from part (a) by h , justify the quotient using part (b), and prove

$$\lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} = \frac{f'(x_0)}{g'(x_0)}.$$

Indication. — For part (b), use $g(x_0 + h)/h \rightarrow g'(x_0) \neq 0$ and the fact that a function with a non-zero limit is non-zero sufficiently near the limit point.

Admitted result. **GDQLR** *General Derivative Quotient Limit Rule*. Let $x_0 \in \mathbb{R}$, let $\delta > 0$, put $I = (x_0 - \delta, x_0 + \delta)$, and let $f, g: I \rightarrow \mathbb{R}$ be continuous. Suppose that f and g are differentiable at every $x \in I \setminus \{x_0\}$, that

$$f(x_0) = g(x_0) = 0, \quad g'(x) \neq 0 \quad (x \in I \setminus \{x_0\}),$$

and that there exists $L \in \mathbb{R}$ such that

$$\lim_{x \rightarrow x_0} \frac{f'(x)}{g'(x)} = L.$$

Then $g(x) \neq 0$ throughout some deleted neighbourhood of x_0 and

$$\lim_{x \rightarrow x_0} \frac{f(x)}{g(x)} = L.$$

13)

- a) **SQR** *Sine Quotient Recovery.* Apply Question 12 to $f(x) = \sin x$ and $g(x) = x$ on $\mathbb{R}$ to recover

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1.$$

- b) **CSOQL** *Cosine Second-Order Quotient Limit.* Explain why Question 12 does not apply directly to $(1 - \cos x)/x^2$ at 0. Use the admitted general rule and part (a) to prove

$$\lim_{x \rightarrow 0} \frac{1 - \cos x}{x^2} = \frac{1}{2}.$$

- c) **RDQE** *Repeated Derivative Quotient Evaluation.* Use the admitted general rule and part (b) to prove

$$\lim_{x \rightarrow 0} \frac{x - \sin x}{x^3} = \frac{1}{6}.$$

- d) **HRVA** *Hypothesis Review before Valid Application.* Define $f, g: \mathbb{R} \rightarrow \mathbb{R}$ by $g(x) = x$ for every $x \in \mathbb{R}$, $f(0) = 0$, and

$$f(x) = x^2 \sin(1/x) \quad (x \in \mathbb{R} \setminus \{0\}).$$

Prove directly that $f(x)/g(x) \rightarrow 0$ as $x \rightarrow 0$. Compute $f'(x)/g'(x)$ for $x \neq 0$ and show that this derivative quotient has no limit as $x \rightarrow 0$. Identify the failed hypothesis of the admitted general rule and explain why the original quotient may nevertheless converge.

End of Part I

Part II. — Integration and multivariable differentiation

This part constructs integration from finite sums and then extends the local viewpoint of Part I to functions of several real variables. The elementary algebra of real limits and continuous functions remains available.

A. — Integrals and their evaluation

Left Riemann sums and definite integrals. Let $a, b \in \mathbb{R}$ satisfy $a < b$, and let $f: [a, b] \rightarrow \mathbb{R}$. For each positive integer $N \in \mathbb{N}$, put

$$\Delta x_N = \frac{b-a}{N}, \quad x_{n,N} = a + n\Delta x_N \quad (0 \leq n \leq N),$$

and define the *left Riemann sum*

$$S_N(f; a, b) = \sum_{n=0}^{N-1} f(x_{n,N}) \Delta x_N.$$

For a continuous function f , the finite real limit $\lim_{N \rightarrow \infty} S_N(f; a, b)$ exists by the result stated immediately below; call it the *definite integral* of f over $[a, b]$ and write it as $\int_a^b f(x) dx$. Define $\int_b^a f(x) dx = -\int_a^b f(x) dx$ and $\int_a^a f(x) dx = 0$. The complete theory of Riemann integration is, as in the original notes, part of Analysis I rather than this course; in particular, this paper does not use left sums as a definition for arbitrary discontinuous functions.

*Admitted result. **BRIP** Basic Riemann Integral Properties.* Let $a, b, c \in \mathbb{R}$ satisfy $a \leq c \leq b$. Every continuous real-valued function on $[a, b]$ is Riemann integrable. If $f, g: [a, b] \rightarrow \mathbb{R}$ are continuous and $\alpha, \beta \in \mathbb{R}$, then

$$\int_a^b (\alpha f(x) + \beta g(x)) dx = \alpha \int_a^b f(x) dx + \beta \int_a^b g(x) dx,$$

and $\int_a^b f = \int_a^c f + \int_c^b f$. If $m, M \in \mathbb{R}$ satisfy $m \leq f(x) \leq M$ for every $x \in [a, b]$, then $m(b-a) \leq \int_a^b f(x) dx \leq M(b-a)$.

14) Let $a, b \in \mathbb{R}$ satisfy $a < b$.

- a) **URSP** *Uniform Riemann Sum Partition.* Let $f: [a, b] \rightarrow \mathbb{R}$. Prove that $x_{0,N} = a$, $x_{N,N} = b$, and $x_{n+1,N} - x_{n,N} = \Delta x_N$ for every $N \in \mathbb{N}$ and every integer n with $0 \leq n < N$. Explain why the term indexed by n is the signed area of a rectangle of width Δx_N and height $f(x_{n,N})$.
- b) **LSRF** *Linear Sum Recovers Formula.* Let $f: [a, b] \rightarrow \mathbb{R}$ be defined by $f(x) = x$. Using $\sum_{n=0}^{N-1} n = N(N-1)/2$, compute $S_N(f; a, b)$ and prove that

$$\int_a^b x dx = \frac{b^2 - a^2}{2}.$$

- c) **RSEO** *Rectangle Sum Error Order.* Suppose that $f: [a, b] \rightarrow \mathbb{R}$ is continuous on $[a, b]$ and differentiable on (a, b) , and that there is a real number $M \geq 0$ such that $|f'(x)| \leq M$ for every $x \in (a, b)$. Use the mean-value theorem to prove that, for every $N \in \mathbb{N}$ and every integer n with $0 \leq n < N$,

$$\left| \int_{x_{n,N}}^{x_{n+1,N}} (f(x) - f(x_{n,N})) dx \right| \leq \frac{M}{2} (\Delta x_N)^2.$$

Deduce that

$$\left| \int_a^b f(x) dx - S_N(f; a, b) \right| \leq \frac{M(b-a)}{2} \Delta x_N,$$

which makes precise the $O((\Delta x_N)^2)$ local error and $O(\Delta x_N)$ total error described in the notes.

Accumulation functions and the Fundamental Theorem. Let $a, b \in \mathbb{R}$ satisfy $a < b$, let $f: [a, b] \rightarrow \mathbb{R}$ be continuous, and define $F: [a, b] \rightarrow \mathbb{R}$ by

$$F(x) = \int_a^x f(t) \, dt \quad (x \in [a, b]).$$

At an endpoint, a derivative below means the appropriate one-sided derivative.

15) Let $x \in [a, b]$ and let $h \in \mathbb{R} \setminus \{0\}$ be such that $x + h \in [a, b]$.

a) **IALI** *Integral Accumulation as a Local Increment.* Use additivity and orientation of the integral to prove

$$\frac{F(x+h) - F(x)}{h} = \frac{1}{h} \int_x^{x+h} f(t) \, dt.$$

b) **CAML** *Continuity gives an Average Mean Limit.* Let $\varepsilon \in \mathbb{R}$ satisfy $\varepsilon > 0$. Use continuity of f at x and the integral bound in BRIP to prove that, whenever h is sufficiently close to 0,

$$\left| \frac{1}{h} \int_x^{x+h} f(t) \, dt - f(x) \right| \leq \varepsilon.$$

Treat both signs of h .

c) **FTCD** *Fundamental Theorem Connecting Differentiation.* Deduce that F is continuous on $[a, b]$, differentiable on (a, b) , and satisfies $F'(x) = f(x)$ for every $x \in (a, b)$, with the corresponding one-sided identities at a and b . This is the Fundamental Theorem of Calculus in the form used in the original notes.

Variable limits and antiderivatives. Let $I \subseteq \mathbb{R}$ be an interval and let $f: I \rightarrow \mathbb{R}$ be continuous. An *antiderivative* of f on I is a differentiable function $A: I \rightarrow \mathbb{R}$ such that $A'(x) = f(x)$ for every interior point $x \in I$, with one-sided derivatives at any included endpoint. The notation $\int f(x) \, dx = A(x) + C$ denotes the family of all antiderivatives, where $C \in \mathbb{R}$ is a constant of integration.

16) Let $a, b \in I$ and let $g: J \rightarrow I$ be differentiable on an interval $J \subseteq \mathbb{R}$.

a) **ORL** *Orientation Reversal for Limits.* Prove that

$$\frac{d}{dx} \int_x^b f(t) \, dt = -f(x)$$

for every interior point $x \in I$ by first reversing the limits.

b) **VLDF** *Variable-Limit Derivative Formula.* Use Question 15 and the chain rule from Question 8 to prove

$$\frac{d}{dx} \int_a^{g(x)} f(t) \, dt = f(g(x))g'(x)$$

at every interior point $x \in J$.

c) **ACFC** *Antiderivatives Carry a Free Constant.* Let $A_1, A_2: I \rightarrow \mathbb{R}$ be antiderivatives of f . Use the mean-value theorem to prove that $A_1 - A_2$ is constant on I and hence justify the constant C in the indefinite-integral notation.

17)

a) **ICS** *Integral Chain Substitution.* Let $I, J \subseteq \mathbb{R}$ be intervals, let $g: I \rightarrow J$ be differentiable, and let $f: J \rightarrow \mathbb{R}$ be continuous. If $A: J \rightarrow \mathbb{R}$ is an antiderivative of f , use the chain rule to prove

$$\int f(g(x))g'(x) \, dx = A(g(x)) + C.$$

Explain the conventional notation $u = g(x)$ and $du = g'(x) \, dx$.

- b) **QSSE** *Quadratic Square root Substitution Example.* On any open interval on which $x - x^2 > 0$, evaluate

$$\int \frac{1 - 2x}{\sqrt{x - x^2}} dx$$

by using $u = x - x^2$. Differentiate your answer to check it.

- c) **STSI** *Shifted Trigonometric Substitution Integral.* On the interval $(0, 2)$, put $x - 1 = \sin \theta$ with $\theta \in (-\pi/2, \pi/2)$. Derive

$$\int \sqrt{2x - x^2} dx = \frac{1}{2} \arcsin(x - 1) + \frac{1}{2}(x - 1)\sqrt{2x - x^2} + C.$$

- d) **HSCP** *Hyperbolic Substitution Choice Practice.* Match each expression $\sqrt{1 - x^2}$, $1 + x^2$, $\sqrt{x^2 - 1}$, $\sqrt{1 + x^2}$, and $1 - x^2$ with a useful substitution chosen from

$$x = \sin \theta, \quad x = \tan \theta, \quad x = \cosh u, \quad x = \sinh u, \quad x = \tanh u,$$

and state the identity that makes each choice useful. Specify a real interval on which each substitution is valid and one-to-one.

- 18) Let $I \subseteq \mathbb{R}$ be an interval and let $u, v: I \rightarrow \mathbb{R}$ be continuously differentiable.

- a) **IBPF** *Integration By-Parts Formula.* Integrate the product-rule identity $(uv)' = u'v + uv'$ and prove

$$\int_a^b u(x)v'(x) dx = [u(x)v(x)]_a^b - \int_a^b v(x)u'(x) dx$$

for every $a, b \in I$. State the corresponding indefinite formula.

- b) **EII** *Exponential Improper Integral.* For a continuous function $q: [0, \infty) \rightarrow \mathbb{R}$, define $\int_0^\infty q(x) dx$ to be $\lim_{R \rightarrow \infty} \int_0^R q(x) dx$ when this finite limit exists. Use integration by parts and the limit $Re^{-R} \rightarrow 0$ to prove

$$\int_0^\infty xe^{-x} dx = 1.$$

- c) **LAI** *Logarithm Antiderivative Identity.* Use integration by parts to prove, for every $x \in (0, \infty)$,

$$\int \log x dx = x \log x - x + C.$$

B. — Partial and total differentiation

Partial derivatives. Let $U \subseteq \mathbb{R}^2$ be open, let $f: U \rightarrow \mathbb{R}$, and let $(x_0, y_0) \in U$. The *partial derivative with respect to the first variable* is

$$f_x(x_0, y_0) = \frac{\partial f}{\partial x}(x_0, y_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + h, y_0) - f(x_0, y_0)}{h},$$

provided the real limit exists. Define $f_y(x_0, y_0)$ analogously by holding x_0 fixed. Iterated derivatives use the order indicated by the subscripts: $f_{xy} = (f_x)_y$ and $f_{yx} = (f_y)_x$.

- 19) Define $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ by $f(x, y) = x^2 + y^3 + e^{xy^2}$ for every $(x, y) \in \mathbb{R}^2$.

- a) **PDQ** *Partial Derivative Quotients.* Starting with the defining limit, compute $f_x(x, y)$ and $f_y(x, y)$ for every $(x, y) \in \mathbb{R}^2$.
- b) **EMPC** *Exponential Multivariable Partial Computation.* Compute $f_{xx}(x, y)$ and $f_{xy}(x, y)$ for every $(x, y) \in \mathbb{R}^2$.
- c) **MPDN** *Mixed Partial Derivative Notation.* Compute $f_{yx}(x, y)$ and verify directly that it equals $f_{xy}(x, y)$. Explain carefully which differentiation is performed first in each subscript convention.

Admitted result. SCECP Schwarz Commutation for Existing Continuous Partial. Let $U \subseteq \mathbb{R}^2$ be open and let $f: U \rightarrow \mathbb{R}$. If the second partial derivatives f_{xy} and f_{yx} exist and are continuous on U , then $f_{xy}(x, y) = f_{yx}(x, y)$ for every $(x, y) \in U$.

Differentiability and the total differential. Let $U \subseteq \mathbb{R}^2$ be open, let $f: U \rightarrow \mathbb{R}$, and let $(x_0, y_0) \in U$. Say that f is *differentiable at* (x_0, y_0) if there exist $A, B \in \mathbb{R}$ and a function r defined near $(0, 0)$ such that, for all sufficiently small $(h, k) \in \mathbb{R}^2$,

$$f(x_0 + h, y_0 + k) - f(x_0, y_0) = Ah + Bk + r(h, k)$$

and $r(h, k)/(|h| + |k|) \rightarrow 0$ as $(h, k) \rightarrow (0, 0)$ with $(h, k) \neq (0, 0)$. The linear expression

$$df = A dx + B dy$$

is called the *total differential* at (x_0, y_0) . A function is *differentiable on* U if it is differentiable at every point of U . Question 20(a) identifies A and B with the two partial derivatives.

20) Let $f: U \rightarrow \mathbb{R}$ be differentiable at $(x_0, y_0) \in U$.

- a) **RPLI Rectangular Path Linear Increment.** Show from the definition that the coefficients satisfy $A = f_x(x_0, y_0)$ and $B = f_y(x_0, y_0)$. Hence prove

$$\delta f = f_x(x_0, y_0)\delta x + f_y(x_0, y_0)\delta y + o(|\delta x| + |\delta y|).$$

- b) **TDAE Total Differential Approximation Error.** Suppose additionally that f_x and f_y exist near (x_0, y_0) and are continuous there. Split the increment along the rectangular path $(x_0, y_0) \rightarrow (x_0 + \delta x, y_0) \rightarrow (x_0 + \delta x, y_0 + \delta y)$ and use one-variable linear approximations to recover the formula in part (a).
- c) **MCR Multivariable Chain Rule.** Let $J \subseteq \mathbb{R}$ be an open interval, let $x, y: J \rightarrow \mathbb{R}$ be differentiable at $t_0 \in J$, and suppose $(x(t), y(t)) \in U$ for every $t \in J$ and $(x(t_0), y(t_0)) = (x_0, y_0)$. Substitute their one-variable linear approximations into that of f to prove

$$\left. \frac{d}{dt} f(x(t), y(t)) \right|_{t=t_0} = f_x(x(t_0), y(t_0))x'(t_0) + f_y(x(t_0), y(t_0))y'(t_0).$$

- d) **CPR Chain rule along a Plane Route.** Now suppose that f is differentiable on U . Let $y: J \rightarrow \mathbb{R}$ be differentiable and suppose $(x, y(x)) \in U$ for every $x \in J$. Deduce

$$\frac{d}{dx} f(x, y(x)) = f_x(x, y(x)) + f_y(x, y(x))y'(x)$$

at every $x \in J$.

21) Let $V = (0, \infty) \times \mathbb{R}$, define $T: V \rightarrow \mathbb{R}^2$ by $T(r, \theta) = (r \cos \theta, r \sin \theta)$, and let $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ be differentiable. Put $F = f \circ T$.

- a) **PCC Polar Coordinate Chain rule.** Apply Question 20 with one variable fixed at a time to prove

$$F_\theta(r, \theta) = -r \sin \theta f_x(T(r, \theta)) + r \cos \theta f_y(T(r, \theta)).$$

- b) **RAD Radial and Angular Derivatives.** Derive a corresponding expression for $F_r(r, \theta)$. For $(x, y) = T(r, \theta)$, solve the resulting two linear equations for $f_x(x, y)$ and $f_y(x, y)$ in terms of $F_r(r, \theta)$ and $F_\theta(r, \theta)$.

Differentiability in three variables. Let $U \subseteq \mathbb{R}^3$ be open, let $F: U \rightarrow \mathbb{R}$, and let $(x_0, y_0, z_0) \in U$. Say that F is *differentiable at* (x_0, y_0, z_0) if there exist $A, B, C \in \mathbb{R}$ and a function r defined near $(0, 0, 0)$ such that, for all sufficiently small $(h, k, \ell) \in \mathbb{R}^3$,

$$F(x_0 + h, y_0 + k, z_0 + \ell) - F(x_0, y_0, z_0) = Ah + Bk + C\ell + r(h, k, \ell),$$

where

$$\frac{r(h, k, \ell)}{|h| + |k| + |\ell|} \longrightarrow 0 \quad \text{as } (h, k, \ell) \rightarrow (0, 0, 0),$$

through non-zero triples. It is differentiable on U if it is differentiable at every point of U . As in two variables, setting two increments equal to zero shows that $A = F_x$, $B = F_y$, and $C = F_z$ at the point in question.

Implicit differentiation. Let $U \subseteq \mathbb{R}^3$ be open, let $F: U \rightarrow \mathbb{R}$ be differentiable, let $c \in \mathbb{R}$, and let $D \subseteq \mathbb{R}^2$ be open. Suppose that a differentiable function $z: D \rightarrow \mathbb{R}$ satisfies $(x, y, z(x, y)) \in U$ and

$$F(x, y, z(x, y)) = c \quad ((x, y) \in D).$$

Thus this equation defines the graph of z as part of the level surface of F with value c .

22)

- a) **IDCE** *Implicit Differentiation Computation Example.* Let $F: \mathbb{R}^3 \rightarrow \mathbb{R}$ be given by $F(x, y, z) = xy^2 + yz^2 + xz^5$. At a point on the level surface $F = 5$ where z is locally a differentiable function of (x, y) , differentiate with respect to x while holding y fixed and prove that, whenever $2yz + 5xz^4 \neq 0$,

$$z_x = -\frac{y^2 + z^5}{2yz + 5xz^4}.$$

- b) **IFD** *Implicit Formula Derivation.* In the general setting above, fix y and use the three-variable linear approximation of F , together with the one-variable linear approximation of $x \mapsto z(x, y)$, to prove that wherever $F_z(x, y, z(x, y)) \neq 0$,

$$z_x = -\frac{F_x}{F_z}.$$

Derive similarly

$$z_y = -\frac{F_y}{F_z},$$

with every partial derivative evaluated at $(x, y, z(x, y))$.

- c) **NZCI** *Non-Zero Condition Interpretation.* Explain why division by F_z requires the stated non-zero condition. Give an example of a differentiable $F: \mathbb{R}^2 \rightarrow \mathbb{R}$ and a level set $F(x, z) = c$ that cannot be represented as a differentiable graph $z = z(x)$ near a point where $F_z = 0$.

C. — Differentiation under an integral sign

Admitted result. **CII** *Continuous Integral Interchange.* Let $a, b, c_0, c_1 \in \mathbb{R}$ satisfy $a < b$ and $c_0 < c_1$. If $f: [a, b] \times [c_0, c_1] \rightarrow \mathbb{R}$ and its partial derivative f_c are continuous, then the function $I: [c_0, c_1] \rightarrow \mathbb{R}$ defined by $I(c) = \int_a^b f(y, c) dy$ is differentiable on (c_0, c_1) and

$$I'(c) = \int_a^b f_c(y, c) dy \quad (c \in (c_0, c_1)),$$

with the corresponding one-sided derivatives at the endpoints.

23) Let $J \subseteq \mathbb{R}$ be an open interval, let $a, b, c: J \rightarrow \mathbb{R}$ be differentiable, and suppose $a(x) < b(x)$ for every $x \in J$. Let $W \subseteq \mathbb{R}^2$ be an open set containing every $(y, c(x))$ with $x \in J$ and $a(x) \leq y \leq b(x)$, and let $f: W \rightarrow \mathbb{R}$ have continuous partial derivatives.

- a) **FPDI** *Fixed-limit Parameter Differentiation under Integration.* For fixed real numbers $\alpha < \beta$, define $K(c) = \int_\alpha^\beta f(y, c) dy$ whenever the segment of integration lies in W . State the derivative $K'(c)$ supplied by CII and explain why continuity is sufficient for the admitted interchange.

- b) **CULR** *Changing Upper Limit Rule.* Fix $x_0 \in J$. By continuity, choose $d \in \mathbb{R}$ and a neighbourhood of x_0 on which $a(x) < d < b(x)$. Define, for arguments near the values in question, $H(u, v) = \int_d^u f(y, v) dy$. Compute H_u and H_v , and apply the multivariable chain rule to $H(b(x), c(x)) - H(a(x), c(x))$.
- c) **GLE** *General Leibniz Expression.* Deduce the Leibniz integral rule

$$\begin{aligned} \frac{d}{dx} \int_{a(x)}^{b(x)} f(y, c(x)) dy &= f(b(x), c(x))b'(x) - f(a(x), c(x))a'(x) \\ &\quad + c'(x) \int_{a(x)}^{b(x)} f_c(y, c(x)) dy. \end{aligned}$$

- d) **PGIE** *Parameterized Gaussian Integral Examples.* For every $\lambda \in (0, \infty)$, define

$$I(\lambda) = \int_0^1 e^{-\lambda y^2} dy, \quad J(\lambda) = \int_0^\lambda e^{-\lambda y^2} dy.$$

Use the Leibniz rule to prove

$$I'(\lambda) = - \int_0^1 y^2 e^{-\lambda y^2} dy, \quad J'(\lambda) = e^{-\lambda^3} - \int_0^\lambda y^2 e^{-\lambda y^2} dy.$$

End of Part II

Part III. — First-order linear differential equations

This part explains the special role of exponentials, introduces the precise classification of first-order equations, and develops homogeneous, inhomogeneous, and variable-coefficient solution methods.

A. — Exponentials as eigenfunctions

Real exponential and logarithm. For each real number $a > 0$, the *exponential function with base a* is the function $E_a: \mathbb{R} \rightarrow (0, \infty)$ defined by $E_a(x) = a^x$. The *natural exponential* is the unique differentiable function $\exp: \mathbb{R} \rightarrow (0, \infty)$ satisfying

$$\exp'(x) = \exp(x) \quad (x \in \mathbb{R}), \quad \exp(0) = 1.$$

Write $e = \exp(1)$ and $e^x = \exp(x)$. The natural exponential is strictly increasing and maps $\mathbb{R}$ onto $(0, \infty)$; its inverse is the *natural logarithm* $\log: (0, \infty) \rightarrow \mathbb{R}$, also written $\ln$. These existence and inverse-function facts belong to the elementary calculus background of the original notes.

24) Let $a \in (0, \infty)$.

- a) **BPDP** *Base Power Derivative Proportionality.* Starting from the derivative definition and the identity $a^{x+h} = a^x a^h$, prove that

$$E'_a(x) = \lambda_a E_a(x), \quad \lambda_a = \lim_{h \rightarrow 0} \frac{a^h - 1}{h},$$

for every $x \in \mathbb{R}$. Explain why λ_a depends on a but not on x .

- b) **NECL** *Natural Exponential Characterizing Limit.* Apply part (a) with $a = e$ and use $\exp'(0) = 1$ to prove $\lambda_e = 1$.
- c) **GENE** *General Exponential through Natural Exponentiation.* From $a = e^{\log a}$, derive

$$a^x = e^{x \log a} \quad (x \in \mathbb{R}),$$

and use the chain rule to prove $\lambda_a = \log a$.

Complex-valued differentiation and eigenfunctions. Let $I \subseteq \mathbb{R}$ be an open interval. A function $f: I \rightarrow \mathbb{C}$ is differentiable at $x \in I$ when the difference quotient has a limit in $\mathbb{C}$; equivalently, its real and imaginary parts are differentiable there. For $\lambda = u + iv \in \mathbb{C}$, where $u, v \in \mathbb{R}$, and $x \in \mathbb{R}$, define

$$e^{\lambda x} = e^{ux} (\cos(vx) + i \sin(vx)).$$

A non-zero differentiable function $f: I \rightarrow \mathbb{C}$ is an *eigenfunction of the differential operator $\frac{d}{dx}$* with *eigenvalue $\lambda \in \mathbb{C}$* when $f'(x) = \lambda f(x)$ for every $x \in I$. Here non-zero means not identically zero.

25) Let $I \subseteq \mathbb{R}$ be a non-empty open interval and let $\lambda \in \mathbb{C}$.

- a) **CEDE** *Complex Exponential Derivative Eigenvalue.* Differentiate the displayed real-and-imaginary-part definition to prove

$$\frac{d}{dx} e^{\lambda x} = \lambda e^{\lambda x} \quad (x \in \mathbb{R}).$$

- b) **OECG** *Only Exponential Candidates are General.* Let $f: I \rightarrow \mathbb{C}$ be differentiable and satisfy $f' = \lambda f$. Define $g(x) = f(x)e^{-\lambda x}$, prove $g'(x) = 0$, and deduce that

$$f(x) = C e^{\lambda x} \quad (x \in I)$$

for some $C \in \mathbb{C}$. Include the zero solution by allowing $C = 0$.

- c) **ICVS** *Initial Condition Value Selection.* Given $x_0 \in I$ and $y_0 \in \mathbb{C}$, determine the unique solution of $f' = \lambda f$ on I satisfying $f(x_0) = y_0$.

B. — Homogeneous equations and two limiting constructions

Classification of first-order linear equations. Let $I \subseteq \mathbb{R}$ be an interval and let $a, b, c: I \rightarrow \mathbb{R}$. An equation for an unknown differentiable function $y: I \rightarrow \mathbb{R}$ is *first-order linear* if it has the form

$$a(x)y'(x) + b(x)y(x) = c(x) \quad (x \in I),$$

that is, if y and y' occur only to the first power and are not multiplied together. It is *homogeneous* if $c(x) = 0$ for every $x \in I$, equivalently if $y = 0$ is a solution, and *inhomogeneous*, or *forced*, otherwise; c is then the *forcing term*. It has *constant coefficients* if a and b are constant. An *initial condition* specifies a value $y(x_0) = y_0$ for some $x_0 \in I$ and $y_0 \in \mathbb{R}$. The same algebraic condition imposed at an endpoint is also called a boundary condition in the original notes.

26) Assume in this question that $p: I \rightarrow \mathbb{R}$ is nonconstant and that $f: I \rightarrow \mathbb{R}$ is not identically zero.

- a) **LHC** *Linear Homogeneous Classification.* Classify each equation

$$y' + p(x)y = 0, \quad y' + p(x)y = f(x), \quad 5y' - 3y = 0, \quad yy' = 1$$

as linear or nonlinear, homogeneous or inhomogeneous, and constant-coefficient or variable-coefficient whenever those terms apply. The functions p and f are the prescribed functions above.

- b) **CAER** *Characteristic Ansatz Equation Reduction.* For the equation $5y' - 3y = 0$ on $\mathbb{R}$, substitute the ansatz $y(x) = e^{mx}$ with $m \in \mathbb{R}$ and derive the algebraic equation $5m - 3 = 0$.
- c) **GSHS** *General Solution of a Homogeneous Scalar equation.* Rewrite the equation as $y' = (3/5)y$ and apply Question 25 to prove that every real-valued solution is

$$y(x) = Ae^{3x/5} \quad (x \in \mathbb{R})$$

for a unique $A \in \mathbb{R}$.

- d) **ICUS** *Initial Condition Uniquely Selects.* Let $x_0, y_0 \in \mathbb{R}$. Find the value of A for which the solution in part (c) satisfies $y(x_0) = y_0$.

Admitted result. **ELC** *Euler Limit Characterization.* For every $z \in \mathbb{R}$,

$$\lim_{n \rightarrow \infty} \left(1 + \frac{z}{n}\right)^n = e^z,$$

where the expression is considered for all sufficiently large positive integers n so that $1 + z/n > 0$.

27) Let $y_{\text{in}} \in \mathbb{R}$ and let $h \in (0, \infty)$. Put $x_n = nh$ for every $n \in \mathbb{N}_0$, and let $(y_n)_{n \in \mathbb{N}_0}$ be a real sequence with $y_0 = y_{\text{in}}$. Approximate the equation $5y' - 3y = 0$ at x_n by replacing $y'(x_n)$ with $(y_{n+1} - y_n)/h$.

- a) **FED** *Forward Euler Discretization.* Derive the recurrence

$$y_{n+1} = \left(1 + \frac{3h}{5}\right) y_n \quad (n \in \mathbb{N}_0).$$

- b) **RPF** *Recurrence Product Formula.* Prove by induction that the unique sequence with initial value y_{in} is

$$y_n = \left(1 + \frac{3h}{5}\right)^n y_{\text{in}} \quad (n \in \mathbb{N}_0).$$

- c) **CEL** *Continuous Exponential Limit.* Fix $x \in (0, \infty)$. For every $N \in \mathbb{N}$, put $h_N = x/N$ and let $(y_j^{(N)})_{j \in \mathbb{N}_0}$ be the sequence from parts (a) and (b) with step h_N and initial value $y_0^{(N)} = y_{\text{in}}$. Prove that its value at the endpoint $x_N = Nh_N = x$ is

$$y_N^{(N)} = y_{\text{in}} \left(1 + \frac{3x}{5N} \right)^N,$$

and use ELC to show that $y_N^{(N)} \rightarrow y_{\text{in}} e^{3x/5}$ as $N \rightarrow \infty$.

A formal power-series method. For the next question, a *formal power series* is an expression $\sum_{n=0}^{\infty} a_n x^n$ manipulated by its coefficients; convergence is not asserted. Define its formal derivative by

$$\left(\sum_{n=0}^{\infty} a_n x^n \right)' = \sum_{n=0}^{\infty} (n+1) a_{n+1} x^n.$$

Two formal power series are equal precisely when their coefficients of every power x^n , $n \in \mathbb{N}_0$, are equal.

Admitted result. EPSI Exponential Power Series Identity. For every $z \in \mathbb{R}$, the real series

$$\sum_{n=0}^{\infty} \frac{z^n}{n!}$$

converges and has sum e^z .

28) Seek a formal solution $y = \sum_{n=0}^{\infty} a_n x^n$ of $5y' - 3y = 0$, where $(a_n)_{n \in \mathbb{N}_0}$ is a sequence in $\mathbb{R}$.

- a) **VPSC** *Vanishing Power Series Coefficients.* Substitute the series and show that equality to the zero series is equivalent to

$$5(n+1)a_{n+1} - 3a_n = 0 \quad (n \in \mathbb{N}_0).$$

- b) **CRC** *Coefficient Recurrence Computation.* Prove by induction that

$$a_n = \frac{1}{n!} \left(\frac{3}{5} \right)^n a_0 \quad (n \in \mathbb{N}_0),$$

using $0! = 1$ and $(n+1)! = (n+1)n!$.

- c) **EPSR** *Exponential Power Series Recovery.* Using EPSI, recover the actual solution $y(x) = a_0 e^{3x/5}$ for every $x \in \mathbb{R}$.

C. — Forcing, superposition, and integrating factors

Particular solutions and complementary functions. Let $I \subseteq \mathbb{R}$ be an interval and let L be a linear differential operator on differentiable functions $I \rightarrow \mathbb{R}$, meaning that

$$L[\alpha u + \beta v] = \alpha L[u] + \beta L[v]$$

for all differentiable $u, v: I \rightarrow \mathbb{R}$ and all $\alpha, \beta \in \mathbb{R}$. For $L[y] = f$, a *particular solution* or *particular integral* is one specific solution y_p ; the *complementary function* y_c is the general solution of $L[y_c] = 0$.

29) Let $f: I \rightarrow \mathbb{R}$ and suppose $y_p: I \rightarrow \mathbb{R}$ satisfies $L[y_p] = f$.

- a) **LODS** *Linear Operator Distribution over Sums.* If $y_c: I \rightarrow \mathbb{R}$ satisfies $L[y_c] = 0$, use linearity to prove $L[y_p + y_c] = f$.
- b) **DHS** *Difference is a Homogeneous Solution.* If $y: I \rightarrow \mathbb{R}$ is any solution of $L[y] = f$, prove that $L[y - y_p] = 0$.
- c) **GSFS** *General Solution Follows Superposition.* Deduce that the complete solution set of $L[y] = f$ is

$$\{y_p + y_c : L[y_c] = 0\}.$$

30) Consider the equation $5y' - 3y = 10$ on $\mathbb{R}$.

- a) **CEP** *Constant Equilibrium Particular solution.* Seek a constant particular solution and prove that $y_p = -10/3$ is the unique constant solution.
- b) **FCGS** *Forced Constant General Solution.* Use Questions 26 and 29 to prove that every solution is

$$y(x) = -\frac{10}{3} + Ae^{3x/5} \quad (x \in \mathbb{R})$$

for a unique $A \in \mathbb{R}$.

- c) **BCAF** *Boundary Condition Applied to the Full solution.* Let $x_0, y_0 \in \mathbb{R}$. Determine A from $y(x_0) = y_0$ and explain why imposing the condition only on the complementary term gives an incorrect answer in general.

31) Let time satisfy $t \in [0, \infty)$. Let $a, b: [0, \infty) \rightarrow \mathbb{R}$ be differentiable abundance functions, let $a_0 \in [0, \infty)$, and let $k_a, k_b \in (0, \infty)$ satisfy $k_a \neq k_b$. Suppose

$$a' = -k_a a, \quad b' = k_a a - k_b b, \quad a(0) = a_0, \quad b(0) = 0.$$

- a) **PADS** *Parent Abundance Decay Solution.* Solve the first initial-value problem and obtain $a(t) = a_0 e^{-k_a t}$ for every $t \geq 0$.
- b) **DFES** *Daughter Forced Exponential Solution.* Substitute part (a) into the equation for b , seek a particular solution $Ce^{-k_a t}$, and determine C .
- c) **ICFR** *Initial Condition Fixes the Response.* Add the complementary solution $De^{-k_b t}$, impose $b(0) = 0$, and prove

$$b(t) = \frac{k_a a_0}{k_b - k_a} (e^{-k_a t} - e^{-k_b t}) \quad (t \geq 0).$$

- d) **IRFC** *Isotope Ratio Formula for Chronology.* Assuming $a_0 > 0$, calculate $b(t)/a(t)$ and explain how a measured ratio, together with known k_a and k_b , gives an equation for the age t .

Standard form and integrating factor. Let $I \subseteq \mathbb{R}$ be an open interval and let $a, b, c: I \rightarrow \mathbb{R}$ be continuous, with $a(x) \neq 0$ for every $x \in I$. Dividing $a(x)y' + b(x)y = c(x)$ by $a(x)$ gives the *standard form*

$$y' + p(x)y = f(x), \quad p = \frac{b}{a}, \quad f = \frac{c}{a}.$$

Fix $x_0 \in I$. The positive function

$$\mu(x) = \exp \left(\int_{x_0}^x p(s) \, ds \right) \quad (x \in I)$$

is called an *integrating factor*.

32)

- a) **IFDE** *Integrating Factor Differential Equation.* Use Question 15 and the chain rule to prove $\mu'(x) = p(x)\mu(x)$ for every $x \in I$.
- b) **PEDF** *Product Exact Derivative Form.* Multiply the standard equation by μ and use the product rule to prove

$$(\mu y)' = \mu f.$$

- c) **GIFF** *General Integrating Factor Formula.* Integrate between x_0 and $x \in I$ and prove that every solution is

$$y(x) = \mu(x)^{-1} \left(C + \int_{x_0}^x \mu(s)f(s) \, ds \right)$$

for a unique $C \in \mathbb{R}$. Prove the converse by differentiation.

33) Consider

$$xy'(x) + (1 - x)y(x) = 1$$

on an interval not containing 0.

- a) **SFDD** *Standard Form Domain Decision.* Divide by x and identify $p(x) = 1/x - 1$ and $f(x) = 1/x$. Explain why the intervals $(0, \infty)$ and $(-\infty, 0)$ must be treated separately.
- b) **IFC** *Integrating Factor Computation.* On $(0, \infty)$, compute an integrating factor $\mu(x) = xe^{-x}$.
- c) **GSBE** *General Solution Before the Endpoint.* Apply Question 32 to prove that every solution on $(0, \infty)$ has the form

$$y(x) = \frac{Ce^x - 1}{x}$$

for some $C \in \mathbb{R}$.

- d) **FBC** *Finite Boundary Condition.* Determine the unique $C \in \mathbb{R}$ for which $y(x)$ has a finite limit as $x \rightarrow 0^+$, and use Question 12 to compute that limit. Explain why this is a boundary regularity condition rather than an initial condition imposed within $(0, \infty)$.

End of Part III

Part IV. — Nonlinear first-order equations and discrete dynamics

This part develops two solvable nonlinear classes and then studies evolution geometrically. Every explicit formula is accompanied by its domain conditions, since division during separation can otherwise discard or add solutions.

A. — Separable and exact equations

Differential form and separation. Let $I, J \subseteq \mathbb{R}$ be intervals, let $D \subseteq \mathbb{R}^2$ be open, and let $P, Q: D \rightarrow \mathbb{R}$ be continuous. For a differentiable function $y: I \rightarrow \mathbb{R}$ whose graph lies in D , the notation

$$P(x, y) dx + Q(x, y) dy = 0$$

means $P(x, y(x)) + Q(x, y(x))y'(x) = 0$ for every $x \in I$. The equation is *separable* on a region if it can be written $q(y) dy = p(x) dx$ for functions of one variable $q: J \rightarrow \mathbb{R}$ and $p: I \rightarrow \mathbb{R}$.

34) Let $I, J \subseteq \mathbb{R}$ be open intervals, let $p: I \rightarrow \mathbb{R}$ and $q: J \rightarrow \mathbb{R}$ be continuous, and let $P_0: I \rightarrow \mathbb{R}$ and $Q_0: J \rightarrow \mathbb{R}$ be antiderivatives of p and q , respectively.

- a) **SDF** *Separable Differential Form.* Prove that a differentiable function $y: I \rightarrow J$ satisfies $q(y(x))y'(x) = p(x)$ for every $x \in I$ if and only if

$$\frac{d}{dx}(Q_0(y(x)) - P_0(x)) = 0 \quad (x \in I).$$

- b) **IISC** *Implicit Integral Solution Criterion.* Deduce that every solution satisfies

$$Q_0(y(x)) = P_0(x) + C \quad (x \in I)$$

for some $C \in \mathbb{R}$, and prove conversely that every differentiable $y: I \rightarrow J$ satisfying this identity solves the equation.

- c) **NSE** *Nonlinear Separation Example.* Let $I \subseteq \mathbb{R}$ be an open interval on which $x^2 \neq 3$, and suppose $y: I \rightarrow \mathbb{R}$ is differentiable and satisfies

$$(x^2 y - 3y)y' - 2xy^2 = 4x.$$

Rearrange without dividing by y to obtain

$$\frac{yy'}{2 + y^2} = \frac{2x}{x^2 - 3},$$

and derive

$$\frac{1}{2} \log(2 + y(x)^2) = \log|x^2 - 3| + C.$$

- d) **DRC** *Domain Restriction Check.* Exponentiate the preceding relation to obtain $\sqrt{2 + y(x)^2} = A|x^2 - 3|$ for some $A > 0$. Differentiate this relation to check the original equation. Explain why $x^2 \neq 3$ is required and why no restriction $y \neq 0$ is needed, since $2 + y^2 > 0$.

Exact equations and simple connectivity. Here a *domain* is a non-empty open connected subset $D \subseteq \mathbb{R}^2$. Let $P, Q: D \rightarrow \mathbb{R}$. The differential form $P dx + Q dy$ is *exact* on D if there is a differentiable function $\Phi: D \rightarrow \mathbb{R}$ satisfying $\Phi_x = P$ and $\Phi_y = Q$ throughout D ; Φ is called a *potential*, and the solution curves are then the level sets $\Phi(x, y) = C$ for $C \in \mathbb{R}$. A domain $D \subseteq \mathbb{R}^2$ is *simply connected* if every continuous closed curve in D can be continuously contracted to a point through curves lying in D . A disc is simply connected, whereas an annulus is not. In three dimensions a sphere is simply connected, whereas a torus is not.

35) Assume $P, Q: D \rightarrow \mathbb{R}$ have continuous first partial derivatives.

- a) **EDC** *Exact Differential Components.* Suppose $P dx + Q dy = d\Phi$ for a function $\Phi: D \rightarrow \mathbb{R}$ having continuous second partial derivatives. Identify P and Q with partial derivatives of Φ .
- b) **MEC** *Mixed Equality Condition.* Apply SCECP to prove the necessary condition

$$P_y(x, y) = Q_x(x, y) \quad ((x, y) \in D).$$

- c) **HDO** *Holed Domain Obstruction.* On $D = \mathbb{R}^2 \setminus \{(0, 0)\}$, define

$$P(x, y) = -\frac{y}{x^2 + y^2}, \quad Q(x, y) = \frac{x}{x^2 + y^2}.$$

Verify $P_y = Q_x$. Parameterize the unit circle by $(x(t), y(t)) = (\cos t, \sin t)$ for $t \in [0, 2\pi]$ and show that a potential would have to increase by 2π around this closed curve. Deduce that equality of mixed derivatives is not sufficient on every domain.

Admitted result. **SCES** *Simply-Connected Exactness Sufficiency.* Let $D \subseteq \mathbb{R}^2$ be an open simply-connected domain, and let $P, Q: D \rightarrow \mathbb{R}$ have continuous first partial derivatives. If $P_y = Q_x$ throughout D , then there exists a single-valued differentiable function $\Phi: D \rightarrow \mathbb{R}$ satisfying $\Phi_x = P$ and $\Phi_y = Q$.

36) On $\mathbb{R}^2$, put $P(x, y) = 2x - 3y^2$ and $Q(x, y) = 6y^2 - 6xy$.

- a) **EVC** *Exactness Verification Calculation.* Compute P_y and Q_x , and use SCES to establish exactness.
- b) **PRF** *Potential Reconstruction Function.* Integrate $\Phi_x = P$ with respect to x and write $\Phi(x, y) = x^2 - 3xy^2 + h(y)$. Use $\Phi_y = Q$ to determine h up to an additive constant.
- c) **ILS** *Implicit Level Solutions.* Prove that the solution curves of

$$6y(y - x)y' + 2x - 3y^2 = 0$$

lie on the level sets

$$x^2 - 3xy^2 + 2y^3 = C,$$

where $C \in \mathbb{R}$, and verify the statement by differentiating along a differentiable branch $y = y(x)$.

B. — Trajectories, isoclines, and stability

Solution curves and isoclines. Let $I \subseteq \mathbb{R}$ be an interval, let $D \subseteq \mathbb{R}^2$, and let $f: D \rightarrow \mathbb{R}$. A differentiable function $y: I \rightarrow \mathbb{R}$ whose graph lies in D and which solves $y' = f(t, y)$ determines a *solution curve*, or *trajectory*, $t \mapsto (t, y(t))$. For $d \in \mathbb{R}$, an *isocline of slope d* is the set of points $(t, y) \in D$ satisfying $f(t, y) = d$.

37) Let $I \subseteq \mathbb{R}$ be an interval and let $y: I \rightarrow \mathbb{R}$ be a differentiable solution of $y' = t(1 - y^2)$.

- a) **EFSC** *Explicit Family of Solution Curves.* On any subinterval where $|y| \neq 1$, separate variables and use

$$\frac{1}{1 - y^2} = \frac{1}{2} \left(\frac{1}{1 + y} + \frac{1}{1 - y} \right)$$

to derive

$$\frac{1}{2} \log \left| \frac{1 + y}{1 - y} \right| = \frac{t^2}{2} + C.$$

Show how this gives $y(t) = (A - e^{-t^2}) / (A + e^{-t^2})$ for some $A \in \mathbb{R} \setminus \{0\}$, wherever the denominator is non-zero.

- b) **MSS Missing Stationary Solutions.** Determine the constant solutions lost when dividing by $1 - y^2$ and check them directly in the differential equation.
- c) **QSP Qualitative Slope Partition.** For $t > 0$, determine the sign of y' in each of the regions $y < -1$, $-1 < y < 1$, and $y > 1$. Show also that every solution has a horizontal tangent at $t = 0$ when $0 \in I$.
- d) **ICD Isocline Curve Derivation.** For $d \in \mathbb{R}$, derive the isocline equation $y^2 = 1 - d/t$ when $t \neq 0$, and state the condition under which its right side is non-negative. Treat $t = 0$ separately. Explain how several such curves give a slope field from which trajectories can be sketched.

Equilibria and the meaning of stability. For $y' = f(t, y)$, an *equilibrium* or *fixed point* is a real number a such that $f(t, a) = 0$ for every allowed t ; then $y(t) = a$ is a constant solution. Assume here that the solutions under discussion are defined for every $t \geq 0$. The equilibrium is *stable* if, for every $\epsilon > 0$, there exists $\delta > 0$ such that $|y(0) - a| < \delta$ implies $|y(t) - a| < \epsilon$ for every $t \geq 0$. It is *attracting* if there exists $\delta > 0$ such that $|y(0) - a| < \delta$ implies $y(t) \rightarrow a$ as $t \rightarrow \infty$, and *asymptotically stable* if both properties hold. It is *unstable* if it is not stable. A *linearisation* keeps only terms linear in a small perturbation.

38) Let $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ have a continuous second partial derivative with respect to y , let $a \in \mathbb{R}$, and suppose $f(t, a) = 0$ for every $t \geq 0$.

- a) **EPE Equilibrium Perturbation Expansion.** Let $y: [0, \infty) \rightarrow \mathbb{R}$ be a differentiable solution and define $\varepsilon: [0, \infty) \rightarrow \mathbb{R}$ by $y(t) = a + \varepsilon(t)$. For each fixed $t \geq 0$, first use Taylor expansion in the scalar increment $\eta \rightarrow 0$ to prove

$$f(t, a + \eta) = f_y(t, a)\eta + O(\eta^2).$$

Substitute $\eta = \varepsilon(t)$ whenever the perturbation is small and deduce

$$\varepsilon'(t) = f_y(t, a)\varepsilon(t) + O(\varepsilon(t)^2)$$

with the remainder understood in this local sense.

- b) **LEM Linearized Evolution Model.** Explain that neglecting the quadratic remainder gives the linearised equation $\varepsilon' = f_y(t, a)\varepsilon$.
- c) **CSULE Contrasting Stability of Upper and Lower Equilibria.** Apply this procedure to $f(t, y) = t(1 - y^2)$. At $a = 1$ and $a = -1$ solve the two linearised equations and obtain respectively $\varepsilon(t) = \varepsilon(0)e^{-t^2}$ and $\varepsilon(t) = \varepsilon(0)e^{t^2}$. Use the explicit family and slope information in Question 37 to confirm that 1 is asymptotically stable and -1 is unstable.
- d) **VALE Validity Area of the Linearized Estimate.** Explain why growth of the linearised perturbation predicts local instability but, without the exact argument in part (c), would not by itself prove it or justify following the approximation after $|\varepsilon|$ ceases to be small.

Autonomous scalar equations. An equation $y' = f(y)$ is *autonomous* because its right-hand side has no explicit dependence on time.

Admitted result. **SAUS Scalar Autonomous Uniqueness for Solutions.** Let $J \subseteq \mathbb{R}$ be an open interval and let $f: J \rightarrow \mathbb{R}$ be continuously differentiable. If two differentiable functions on the same time interval solve $y' = f(y)$ and have the same value at one time, then they agree throughout that time interval.

39) Let $J \subseteq \mathbb{R}$ be an open interval, let $f: J \rightarrow \mathbb{R}$ be twice continuously differentiable, and let $a \in J$ satisfy $f(a) = 0$.

- a) **ASDC Autonomous Stability Derivative Criterion.** Show that the linearised perturbation equation is $\varepsilon' = f'(a)\varepsilon$ and solve it. Use continuity of f' to determine the sign of $(y - a)f(y)$ near a , and use SAUS to show that a nonconstant solution

cannot cross a . Deduce asymptotic stability when $f'(a) < 0$ and instability when $f'(a) > 0$.

- b) **NDC** *Non-hyperbolic Derivative Caveat*. Explain why $f'(a) = 0$ makes the linearised equation inconclusive. Compare $y' = -y^3$ and $y' = y^3$ near the common equilibrium $a = 0$ by inspecting their signs.

Indication.— For part (a), continuity of f' gives a punctured interval on which $f(y)$ points towards a when $f'(a) < 0$ and away from a when $f'(a) > 0$. Use SAUS to prevent crossing. A bounded monotone solution has a limit; if that limit were a nearby point different from a , continuity would make its derivative stay bounded away from zero.

40) Let $a_0, b_0, \lambda \in (0, \infty)$ satisfy $a_0 < b_0$. In a dilute solution, let $a, b, c: [0, \infty) \rightarrow [0, \infty)$ be continuous and differentiable for $t > 0$, denoting the amounts of NaOH, HCl, and each reaction product. Suppose $a(0) = a_0$, $b(0) = b_0$, and $c(0) = 0$. Assume the reaction rate is $\lambda a(t)b(t)$ for $t > 0$.

- a) **MBM** *Mass Balance Model*. Use the reaction stoichiometry to obtain $a = a_0 - c$, $b = b_0 - c$, and

$$c' = \lambda(a_0 - c)(b_0 - c).$$

- b) **ECSA** *Equilibria and Chemical Stability Analysis*. Find the equilibria $c = a_0$ and $c = b_0$. Use the sign of the right-hand side, or Question 39, to classify them.
- c) **SCNF** *Separable Chemical Neutralization Formula*. Separate variables, use partial fractions, impose $c(0) = 0$, and derive

$$c(t) = \frac{a_0 b_0 (1 - e^{-(b_0 - a_0)\lambda t})}{b_0 - a_0 e^{-(b_0 - a_0)\lambda t}} \quad (t \geq 0).$$

- d) **PDR** *Physical Domain Restriction*. Prove that $0 \leq c(t) < a_0$ for every finite $t \geq 0$ and that $c(t) \rightarrow a_0$ as $t \rightarrow \infty$. Explain why branches with $c > a_0$ do not describe non-negative reactant amounts.

C. — Continuous and discrete logistic growth

Logistic equation. Let $r, Y \in (0, \infty)$. The *logistic equation* for a population $y: [0, \infty) \rightarrow [0, \infty)$ is

$$y' = ry \left(1 - \frac{y}{Y}\right),$$

where r is the net low-density growth rate and Y is the *carrying capacity*. In the model of the notes, $r = \alpha - \beta > 0$ for per-capita birth and death rates $\alpha, \beta \in [0, \infty)$, and competition adds a death term γy^2 with $\gamma > 0$, giving $Y = r/\gamma$.

41)

- a) **ELG** *Exponential Low-density Growth*. Let $\rho \in \mathbb{R}$ and $y_0 \in [0, \infty)$. Omit competition and solve $y' = \rho y$ with $y(0) = y_0$. Explain why this predicts exponential growth when $\rho > 0$ and exponential decay when $\rho < 0$.
- b) **CPC** *Competition Produces Carrying capacity*. Starting from $(\alpha - \beta)y - \gamma y^2$, factor the right-hand side into logistic form and identify Y .
- c) **LESC** *Logistic Equilibria and Stability Classification*. Find the equilibria 0 and Y , compute the derivative of $f(y) = ry(1 - y/Y)$ at each, and classify them using Question 39.
- d) **LES** *Logistic Explicit Solution*. For $y_0 \in (0, \infty)$, separate variables and prove

$$y(t) = \frac{Y}{1 + (Y/y_0 - 1)e^{-rt}} \quad (t \geq 0).$$

Verify its low-density approximation and its limit as $t \rightarrow \infty$.

Discrete dynamical systems. Let $S \subseteq \mathbb{R}$ and let $F: S \rightarrow S$. A *difference equation*, or *discrete dynamical system*, is a sequence $(x_n)_{n \in \mathbb{N}_0}$ satisfying $x_{n+1} = F(x_n)$ for every $n \in \mathbb{N}_0$. A *fixed point* is $x^* \in S$ with $F(x^*) = x^*$. It is *stable* if, for every $\epsilon > 0$, there exists $\delta > 0$ such that $|x_0 - x^*| < \delta$ implies $|x_n - x^*| < \epsilon$ for every $n \in \mathbb{N}_0$; it is *locally attracting* if all sufficiently close initial values produce $x_n \rightarrow x^*$ as $n \rightarrow \infty$, and it is *unstable* if it is not stable. For $\lambda > 0$, the *logistic map* is the function $F_\lambda: \mathbb{R} \rightarrow \mathbb{R}$ defined by

$$F_\lambda(x) = \lambda x(1 - x), \quad x_{n+1} = F_\lambda(x_n).$$

A *cobweb diagram* alternates vertical segments to the graph of F_λ and horizontal segments to the diagonal $x_{n+1} = x_n$, thereby displaying successive iterates.

42)

- a) **FELM** *Forward Euler Logistic Map.* Let $\Delta t > 0$ and $y_0 \in [0, \infty)$. Apply a forward difference of step Δt to the continuous logistic equation to obtain a recurrence for a real sequence $(y_n)_{n \in \mathbb{N}_0}$ with initial value y_0 . With

$$\lambda = 1 + r\Delta t, \quad x_n = \frac{r\Delta t}{1 + r\Delta t} \frac{y_n}{Y},$$

derive $x_{n+1} = \lambda x_n(1 - x_n)$.

- b) **DFF** *Discrete Fixed Points.* Solve $F_\lambda(x) = x$ and obtain $x = 0$ and, when $\lambda \neq 0$, $x = 1 - 1/\lambda$.
- c) **DLSC** *Discrete Linear Stability Criterion.* Let $J \subseteq \mathbb{R}$ be an open interval, let $F: J \rightarrow J$ be twice differentiable near a fixed point $x^* \in J$. Put $x_n = x^* + \varepsilon_n$ and show $\varepsilon_{n+1} = F'(x^*)\varepsilon_n + O(\varepsilon_n^2)$. Deduce local attraction when $|F'(x^*)| < 1$ and instability when $|F'(x^*)| > 1$.
- d) **PRC** *Parameter Regime Classification.* Compute F'_λ at both fixed points. Show that 0 is attracting for $0 < \lambda < 1$, while $1 - 1/\lambda$ is attracting for $1 < \lambda < 3$. Explain from the sign of $2 - \lambda$ why convergence is locally monotone for $1 < \lambda < 2$ and alternating for $2 < \lambda < 3$.
- e) **CWI** *Cobweb Walk Interpretation.* Describe the cobweb construction from an initial value x_0 . Relate its inward straight or alternating steps to each of the three attracting regimes in part (d).

Indication.— For part (c), when $|F'(x^*)| < 1$, use continuity of F' and the mean-value theorem to find a neighbourhood on which F contracts distances from x^* . When $|F'(x^*)| > 1$, find a neighbourhood on which each non-zero displacement expands until it leaves that neighbourhood.

Periodic orbits. Retain a set $S \subseteq \mathbb{R}$ and a function $F: S \rightarrow S$. A *periodic orbit of least period* $k \in \mathbb{N}$ is a list of distinct points $x_0, \dots, x_{k-1} \in S$ such that $F(x_j) = x_{j+1}$ for every $j \in \mathbb{N}_0$ with $0 \leq j < k-1$, $F(x_{k-1}) = x_0$, and no smaller positive period has this property. It is also called a limit cycle in the terminology of the original notes.

43) Let $F_\lambda(x) = \lambda x(1 - x)$ with $\lambda > 0$.

- a) **TPE** *Two-cycle Polynomial Extraction.* Compute $F_\lambda(F_\lambda(x)) - x$, divide out the factors corresponding to the two fixed points, and show that the remaining quadratic has two distinct real roots precisely when $\lambda > 3$.

Indication.— After removing x and $x - (1 - 1/\lambda)$, the quadratic may be written $\lambda^2 x^2 - \lambda(\lambda + 1)x + (\lambda + 1) = 0$.

- b) **TCS** *Two-Cycle Stability.* If x_1, x_2 are the two roots from part (a), show that the derivative of the second iterate at either point is $F'_\lambda(x_1)F'_\lambda(x_2)$. Compute its absolute value and prove that the two-cycle is attracting exactly when $3 < \lambda < 1 + \sqrt{6}$.

Admitted result. **LMPDB** *Logistic Map Period-Doubling Behaviour.* As λ increases beyond $1 + \sqrt{6}$, the logistic map's attracting two-cycle loses stability and an attracting four-cycle appears. Further increases produce a cascade of period doublings, including cycles of periods 8, 16, $\dots$, followed by parameter ranges with chaotic dynamics. The earlier fixed point and periodic orbits continue to exist after losing stability.

End of Part IV

Part V. — Homogeneous second-order equations and phase space

Second-order equations arise when a model contains acceleration. This part classifies constant-coefficient homogeneous solutions and gives a geometric meaning to the two initial data required by an equation of order two.

Second-order linear equations. Let $I \subseteq \mathbb{R}$ be an interval, let $a, b, c \in \mathbb{R}$ with $a \neq 0$, and let $f: I \rightarrow \mathbb{R}$. A *second-order linear ordinary differential equation with constant coefficients* has the form

$$ay''(x) + by'(x) + cy(x) = f(x) \quad (x \in I),$$

for an unknown twice differentiable function $y: I \rightarrow \mathbb{R}$. It is *homogeneous* when f is the zero function. For an inhomogeneous equation, linearity gives the same two-step strategy as in Part III: find the general complementary function y_c and one particular integral y_p , then write $y = y_c + y_p$, where $y_c, y_p: I \rightarrow \mathbb{R}$.

Admitted result. LIVPEU Linear Initial-Value Problem Existence and Uniqueness. Let $I \subseteq \mathbb{R}$ be an open interval, let $p, q, r: I \rightarrow \mathbb{R}$ be continuous, let $x_0 \in I$, and let $u_0, v_0 \in \mathbb{R}$. There exists a unique twice continuously differentiable function $y: I \rightarrow \mathbb{R}$ satisfying

$$y'' + p(x)y' + q(x)y = r(x) \quad (x \in I), \quad y(x_0) = u_0, \quad y'(x_0) = v_0.$$

More generally, let $n \in \mathbb{N}$, let $I \subseteq \mathbb{R}$ be an open interval, let $a_0, \dots, a_n, r: I \rightarrow \mathbb{R}$ be continuous, and suppose $a_n(x) \neq 0$ for every $x \in I$. For every $x_0 \in I$ and every $(u_0, \dots, u_{n-1}) \in \mathbb{R}^n$, there exists a unique n -times continuously differentiable function $y: I \rightarrow \mathbb{R}$ satisfying

$$\sum_{j=0}^n a_j(x)y^{(j)}(x) = r(x) \quad (x \in I), \quad y^{(j)}(x_0) = u_j \quad (j \in \mathbb{N}_0, 0 \leq j < n).$$

A. — Characteristic roots

44) Let $a, b, c \in \mathbb{R}$ with $a \neq 0$ and seek complex-valued solutions of $ay'' + by' + cy = 0$ on $\mathbb{R}$.

- a) **CEFE Characteristic Equation From an Exponential.** Substitute $y(x) = e^{\lambda x}$ with $\lambda \in \mathbb{C}$ and prove that it is a solution if and only if

$$a\lambda^2 + b\lambda + c = 0.$$

This polynomial equation is called the *characteristic equation*.

- b) **LIDS Linear Independence for Distinct Solutions.** Let $\lambda_1, \lambda_2 \in \mathbb{C}$ be distinct characteristic roots. If $A, B \in \mathbb{C}$ satisfy $Ae^{\lambda_1 x} + Be^{\lambda_2 x} = 0$ for every $x \in \mathbb{R}$, evaluate the identity and its derivative at $x = 0$ and prove $A = B = 0$.
- c) **DRGS Distinct-Root General Solution.** For arbitrary $u_0, v_0 \in \mathbb{C}$, impose the complex initial data $y(0) = u_0$ and $y'(0) = v_0$, and solve uniquely for $A, B \in \mathbb{C}$ in $y = Ae^{\lambda_1 x} + Be^{\lambda_2 x}$. Use LIVPEU, applied to real and imaginary parts, to deduce that this is the general complex-valued solution.
- d) **DRWE Distinct Real-root Worked Example.** Solve $y'' - 5y' + 6y = 0$ on $\mathbb{R}$ and obtain $y = Ae^{2x} + Be^{3x}$, where $A, B \in \mathbb{R}$ for real-valued solutions.

45)

- a) **CSHM Complex Solutions of Harmonic Motion.** Solve the characteristic equation for $y'' + 4y = 0$ and obtain the complex basis e^{2ix}, e^{-2ix} .

- b) **RBCT** *Real Basis through Complex Transformation.* Use $e^{i\theta} = \cos \theta + i \sin \theta$ to show that every real-valued solution can be written uniquely as

$$y(x) = \alpha \cos(2x) + \beta \sin(2x)$$

with $\alpha, \beta \in \mathbb{R}$. Relate these constants to the complex coefficients $A, B \in \mathbb{C}$ under the condition that y is real.

- c) **DRSS** *Detuning Reveals a Second Solution.* For $\varepsilon \in \mathbb{R} \setminus \{0\}$, solve

$$y'' - 4y' + (4 - \varepsilon^2)y = 0.$$

Expand $e^{\pm \varepsilon x}$ at $\varepsilon = 0$ and show how the two combinations

$$\frac{e^{(2+\varepsilon)x} + e^{(2-\varepsilon)x}}{2}, \quad \frac{e^{(2+\varepsilon)x} - e^{(2-\varepsilon)x}}{2\varepsilon}$$

tend respectively to e^{2x} and xe^{2x} .

- d) **RRGS** *Repeated-Root General Solution.* Let $a\lambda^2 + b\lambda + c$ have a repeated root $\lambda_0 \in \mathbb{R}$. Verify directly that $e^{\lambda_0 x}$ and $xe^{\lambda_0 x}$ solve the equation. Show that their initial state vectors at $x = 0$ are $(1, \lambda_0)$ and $(0, 1)$, and hence that their linear combinations realize arbitrary data in $\mathbb{R}^2$. Use LIVPEU to prove that the general real solution is

$$y(x) = (A + Bx)e^{\lambda_0 x}, \quad A, B \in \mathbb{R}.$$

Reduction of order. Given one non-zero solution $y_1: I \rightarrow \mathbb{R}$ of a homogeneous second-order linear equation, *reduction of order* seeks another solution in the form $y_2 = vy_1$, where $v: I \rightarrow \mathbb{R}$ is an unknown twice differentiable function.

46) For $y'' - 4y' + 4y = 0$ on $\mathbb{R}$, take $y_1(x) = e^{2x}$, let $v: \mathbb{R} \rightarrow \mathbb{R}$ be twice differentiable, and set $y_2(x) = v(x)e^{2x}$ for every $x \in \mathbb{R}$.

- a) **ROA** *Reduction of Order Ansatz.* Compute y_2' and y_2'' in terms of v, v' , and v'' .
 b) **REEC** *Repeated Equation Exact Cancellation.* Substitute these expressions and prove that every term except $v''(x)e^{2x}$ cancels.
 c) **SVLV** *Second solution Via a Linear Variation.* Solve $v'' = 0$, discard the component proportional to y_1 , and recover the second solution $y_2(x) = xe^{2x}$. Explain how this confirms the repeated-root formula without a detuning limit.

B. — Phase space and the Wronskian

State vectors and phase space. For a positive integer $n \in \mathbb{N}$, let $I \subseteq \mathbb{R}$ be an interval and consider

$$a_n(x)y^{(n)} + a_{n-1}(x)y^{(n-1)} + \cdots + a_1(x)y' + a_0(x)y = f(x),$$

where $a_0, \dots, a_n, f: I \rightarrow \mathbb{R}$ and $a_n(x) \neq 0$ for every $x \in I$, for an unknown n -times differentiable function $y: I \rightarrow \mathbb{R}$. Its *state vector* or *solution vector* is

$$\mathbf{Y}(x) = (y(x), y'(x), \dots, y^{(n-1)}(x)) \in \mathbb{R}^n.$$

The ambient space $\mathbb{R}^n$ of possible states is the *phase space*; a solution traces a phase-space trajectory.

47)

- a) **SFOS** *State First-Order System.* Solve the displayed differential equation algebraically for $y^{(n)}$ and show that it determines the derivative of every component of $\mathbf{Y}$ from x and $\mathbf{Y}(x)$.

- b) **PID** *Phase Initial Data*. Assume that I is open and that $a_0, \dots, a_n$ and f are continuous on I , and let $x_0 \in I$. Explain why an initial state $\mathbf{Y}(x_0) = (u_0, \dots, u_{n-1}) \in \mathbb{R}^n$ prescribes exactly $y(x_0), \dots, y^{(n-1)}(x_0)$. State the uniqueness conclusion supplied by LIVPEU.
- c) **SHPT** *Simple Harmonic Phase Trajectory*. For $y'' + 4y = 0$ with $y(0) = 1$ and $y'(0) = 0$, find y and prove that

$$\mathbf{Y}(x) = (\cos 2x, -2 \sin 2x)$$

traces the ellipse $y^2 + (y'/2)^2 = 1$ clockwise as x increases from 0.

- d) **SIBT** *Second Independent Basis Trajectory*. Repeat part (c) for $y(0) = 0$ and $y'(0) = 2$. Show that the two initial state vectors $(1, 0)$ and $(0, 2)$ form a basis of $\mathbb{R}^2$ and that the corresponding state vectors remain independent for every $x \in \mathbb{R}$.

Wronskian. Let $I \subseteq \mathbb{R}$ be an interval. For differentiable functions $y_1, y_2: I \rightarrow \mathbb{R}$, their *Wronskian* is

$$W(x) = \begin{vmatrix} y_1(x) & y_2(x) \\ y_1'(x) & y_2'(x) \end{vmatrix} = y_1(x)y_2'(x) - y_2(x)y_1'(x).$$

The functions are *linearly independent* if the identity $Ay_1 + By_2 = 0$ on I , for constants $A, B \in \mathbb{R}$, implies $A = B = 0$.

48) Let $I \subseteq \mathbb{R}$ be an open interval, let $p, q: I \rightarrow \mathbb{R}$ be continuous, and let $y_1, y_2: I \rightarrow \mathbb{R}$ solve $y'' + p(x)y' + q(x)y = 0$ on I .

- a) **WIC** *Wronskian Independence Certificate*. Prove that $W(x_0) \neq 0$ at some $x_0 \in I$ implies linear independence. Conversely, if $W(x_0) = 0$, choose constants A, B , not both zero, so that $Ay_1 + By_2$ and its derivative vanish at x_0 , and use LIVPEU to prove dependence.
- b) **WSH** *Wronskian for Simple Harmonics*. Compute the Wronskian of $\cos 2x$ and $\sin 2x$ and obtain $W(x) = 2$ for every $x \in \mathbb{R}$.
- c) **WRR** *Wronskian for a Repeated Root*. Compute the Wronskian of e^{2x} and xe^{2x} and obtain $W(x) = e^{4x}$ for every $x \in \mathbb{R}$.

49) Let $I \subseteq \mathbb{R}$ be an open interval, let $p, q: I \rightarrow \mathbb{R}$ be continuous, and let $y_1, y_2: I \rightarrow \mathbb{R}$ solve $y'' + p(x)y' + q(x)y = 0$ on I .

- a) **SWI** *Subtracted Wronskian Identity*. Multiply the equation for y_1 by y_2 , multiply that for y_2 by y_1 , subtract, and identify the result in terms of W and W' .
- b) **AWE** *Abel Wronskian Equation*. Prove $W' + pW = 0$ and apply Question 32 to obtain

$$W(x) = W(x_0) \exp \left(- \int_{x_0}^x p(s) \, ds \right)$$

for all $x, x_0 \in I$.

- c) **ZONZ** *Zero Or Never Zero*. Deduce Abel's theorem: the Wronskian is either zero throughout I or non-zero at every point of I . Combine this with Question 48 to explain why independence at one point persists along the whole trajectory.

Part IX will define the coefficient matrix, its trace, and the higher-order Wronskian before formulating the corresponding determinant identity for systems of first-order equations.

End of Part V

Part VI. — Forcing, scaling, recurrences, damping, and impulses

This part completes the constant-coefficient theory and compares differential equations with two related structures: equidimensional equations and linear recurrences. It ends with physical forcing over long and very short time scales.

A. — Particular integrals and resonance

Undetermined coefficients. For constants $a, b, c \in \mathbb{R}$ with $a \neq 0$, write $L[y] = ay'' + by' + cy$. The *method of undetermined coefficients* chooses a trial particular integral whose algebraic form resembles a simple forcing term and determines its coefficients by substitution. Linearity allows separate particular integrals to be added.

50)

- a) **TSL** *Trial Shape List.* Let $m, k \in \mathbb{R}$ and $n \in \mathbb{N}_0$. Explain why useful trial spaces for forcing terms e^{mx} , $\sin(kx)$ or $\cos(kx)$, and a polynomial of degree n are, respectively,

$$Ae^{mx}, \quad A \sin(kx) + B \cos(kx), \quad A_n x^n + \cdots + A_1 x + A_0,$$

provided none of the trial functions already belongs to the complementary solution space.

- b) **PEFC** *Polynomial and Exponential Forcing Computation.* For

$$y'' - 5y' + 6y = 2x + e^{4x},$$

use $y_p = ax + b + ce^{4x}$ and prove $a = 1/3$, $b = 5/18$, and $c = 1/2$.

- c) **CSFS** *Complementary Superposition Forms the Solution.* Determine the complementary function and prove that the general solution is

$$y(x) = Ae^{2x} + Be^{3x} + \frac{1}{3}x + \frac{5}{18} + \frac{1}{2}e^{4x}, \quad A, B \in \mathbb{R}.$$

Explain where initial or boundary conditions must be applied.

Resonance and beating. For a linear equation $L[y] = f$, *resonance* occurs when the forcing f is itself a complementary solution, so the usual same-form trial is annihilated by L . *Beating* is slow amplitude modulation caused by forcing at a frequency close to the natural frequency.

51) Let $\omega_0 \in (0, \infty)$.

- a) **SGRF** *Standard Guess Resonance Failure.* For $y'' + \omega_0^2 y = \sin(\omega_0 t)$, prove that every trial $C \sin(\omega_0 t) + D \cos(\omega_0 t)$ is annihilated by the left-hand operator and hence cannot be a particular integral.
- b) **DNPI** *Detuned Nonresonant Particular Integral.* Let $\omega \in (0, \infty) \setminus \{\omega_0\}$. Verify that

$$y_p(t) = \frac{\sin(\omega t) - \sin(\omega_0 t)}{\omega_0^2 - \omega^2}$$

solves $y'' + \omega_0^2 y = \sin(\omega t)$; explain why subtracting the complementary term does not change the forcing.

- c) **BLS** *Beating Length Scales.* Put $\Delta\omega = \omega_0 - \omega$ and use a product-to-sum identity to express y_p as a fast cosine times a slow sine envelope. Identify the respective time scales $O(1/\omega_0)$ and $O(1/|\Delta\omega|)$ when $|\Delta\omega| \ll \omega_0$.
- d) **RPL** *Resonant Particular Limit.* For each fixed $t \in \mathbb{R}$, take $\omega \rightarrow \omega_0$, using $\sin z/z \rightarrow 1$, and prove

$$y_p(t) \longrightarrow -\frac{t}{2\omega_0} \cos(\omega_0 t).$$

Verify this limiting function directly in the resonant equation and explain why its linearly growing amplitude is the signature of resonance.

- e) **GRM** *General Resonance Multiplier*. Let $a, b, c \in \mathbb{R}$ with $a \neq 0$, let $P(z) = az^2 + bz + c$, and suppose that $\lambda \in \mathbb{C}$ is a simple root of P . For $L[y] = ay'' + by' + cy$, verify

$$L[Axe^{\lambda x}] = A(2a\lambda + b)e^{\lambda x},$$

and choose $A \in \mathbb{C}$ so this expression equals $e^{\lambda x}$. State the analogous real sine-cosine trial multiplied by x when the simple characteristic roots are $\pm i\omega_0$ with $\omega_0 > 0$.

Variation of parameters. Let $I \subseteq \mathbb{R}$ be an open interval, let $p, q, f: I \rightarrow \mathbb{R}$ be continuous, and let $y_1, y_2: I \rightarrow \mathbb{R}$ be linearly independent solutions of $y'' + py' + qy = 0$. Their Wronskian W never vanishes by Question 48. Variation of parameters seeks a particular solution of $y'' + py' + qy = f$ in the form $y_p = uy_1 + vy_2$, where $u, v: I \rightarrow \mathbb{R}$ are continuously differentiable coefficient functions.

52)

- a) **VPA** *Variation Parameter Ansatz*. Differentiate $y_p = uy_1 + vy_2$ and impose the auxiliary condition $u'y_1 + v'y_2 = 0$ to obtain $y'_p = uy'_1 + vy'_2$.
- b) **ADC** *Auxiliary Derivative Constraint*. Differentiate once more, substitute into the inhomogeneous equation, and use the two homogeneous equations to prove $u'y'_1 + v'y'_2 = f$.
- c) **WLS** *Wronskian Linear System*. Write the two equations for (u', v') as

$$\begin{pmatrix} y_1 & y_2 \\ y'_1 & y'_2 \end{pmatrix} \begin{pmatrix} u' \\ v' \end{pmatrix} = \begin{pmatrix} 0 \\ f \end{pmatrix}$$

and solve them to obtain $u' = -y_2 f / W$ and $v' = y_1 f / W$.

- d) **VPIF** *Variation Parameters Integral Formula*. Fix $x_0 \in I$, integrate the preceding derivatives, and prove that

$$y_p(x) = -y_1(x) \int_{x_0}^x \frac{y_2(s)f(s)}{W(s)} ds + y_2(x) \int_{x_0}^x \frac{y_1(s)f(s)}{W(s)} ds$$

is a particular solution. Explain why changing either integration constant only adds a complementary solution.

53) For $y'' + 4y = \sin(2x)$, take $y_1(x) = \sin(2x)$ and $y_2(x) = \cos(2x)$.

- a) **WSEC** *Wronskian Sign and Elementary Coefficients*. Compute $W = -2$ and derive

$$u' = \frac{1}{4} \sin(4x), \quad v' = \frac{1}{4} (\cos(4x) - 1).$$

- b) **VPIE** *Variation Parameter Integration Example*. Integrate, substitute in $uy_1 + vy_2$, and simplify to a particular integral differing from $-(x/4) \cos(2x)$ by at most a complementary function.
- c) **DVA** *Detuning and Variation Agree*. Compare with Question 51 and verify directly that $-(x/4) \cos(2x)$ gives the required forcing.

B. — Equidimensional equations and recurrences

Equidimensional equation. On $(0, \infty)$, a second-order *equidimensional*, *Euler*, or *Cauchy–Euler* equation has the form

$$ax^2y'' + bxy' + cy = f(x),$$

where $a, b, c \in \mathbb{R}$, $a \neq 0$, and $f: (0, \infty) \rightarrow \mathbb{R}$ is continuous, for an unknown twice differentiable function $y: (0, \infty) \rightarrow \mathbb{R}$. The powers of x make every term have the

dimensions of y : if $[x] = T$ and $[y] = L$, then $[x^2 y''] = [xy'] = [y] = L$. The equation is *homogeneous* when f is the zero function. Some texts use “homogeneous equation” as a synonym for “equidimensional equation”; here that phrase always means that the forcing is zero.

54)

- a) **IEPP** *Indicial Equation from a Power Profile*. Substitute $y = x^k$ with $k \in \mathbb{R}$ in the homogeneous equation and derive the *indicial equation*

$$ak(k-1) + bk + c = 0.$$

- b) **DRCF** *Distinct-Root Complementary Form*. If the indicial roots $k_1, k_2 \in \mathbb{R}$ are distinct, verify that $Ax^{k_1} + Bx^{k_2}$ is a solution for all $A, B \in \mathbb{R}$.
- c) **LTCC** *Logarithmic Transformation to Constant Coefficients*. Put $z = \log x$ and $Y(z) = y(e^z)$. Prove

$$xy' = Y_z, \quad x^2 y'' = Y_{zz} - Y_z,$$

and transform the equation into

$$aY_{zz} + (b-a)Y_z + cY = f(e^z).$$

Use LIVPEU to justify that the two functions in part (b) span the homogeneous solution space.

- d) **RRL** *Repeated Root Logarithm*. If the indicial equation has a repeated real root k , use the transformed constant-coefficient equation to prove

$$y_c(x) = (A + B \log x)x^k.$$

- e) **RFR** *Resonant Forcing Response*. Suppose k_1 is a simple indicial root and $f(x)$ is proportional to x^{k_1} . Transform to z and explain why a particular trial $Cx^{k_1} \log x$ is appropriate. State the extra factor required when the indicial root is repeated.

Second-order linear recurrences. Let $a, b, c \in \mathbb{R}$ with $a \neq 0$, and let $f: \mathbb{N}_0 \rightarrow \mathbb{R}$. A second-order linear difference equation with constant coefficients is

$$ay_{n+2} + by_{n+1} + cy_n = f_n \quad (n \in \mathbb{N}_0)$$

for an unknown sequence $y: \mathbb{N}_0 \rightarrow \mathbb{R}$, where $y_n = y(n)$ and $f_n = f(n)$. It is homogeneous when $f_n = 0$ for every $n \in \mathbb{N}_0$. The *shift operator* is $S[y]_n = y_{n+1}$.

55)

- a) **SEGP** *Shift Eigenfunction as a Geometric Progression*. For $k \in \mathbb{R}$, define $g_n = k^n$, using $0^0 = 1$. Show that $S[g]_n = kg_n$ for every $n \in \mathbb{N}_0$.
- b) **DCEC** *Difference Characteristic Equation Calculation*. Substitute $y_n = k^n$ and derive $ak^2 + bk + c = 0$. If its two roots k_1, k_2 are distinct and non-zero, prove that $y_n = Ak_1^n + Bk_2^n$ realizes arbitrary initial data y_0, y_1 and hence is the general homogeneous solution.
- c) **RRCF** *Repeated root Recurrence Complementary Form*. If the characteristic polynomial has a repeated non-zero root k , verify directly that k^n and nk^n are solutions and deduce $y_n = (A + Bn)k^n$.
- d) **DZRH** *Degenerate Zero-Root Handling*. If $c = 0$, factor the recurrence operator and discuss separately the zero characteristic root. Show in particular why the formula in part (c) requires $k \neq 0$ when the zero root is repeated.
- e) **PIG** *Particular Inspection Guesses*. Let $r \in \mathbb{R} \setminus \{0\}$ and $p \in \mathbb{N}_0$. Give suitable trial forms for forcing r^n when r is not a characteristic root and for polynomial forcing of degree p when 1 is not a characteristic root. Explain why a geometric trial resonant with a simple root is multiplied by n , in parallel with Question 51, and state the additional factor required for a repeated root.

56) Let $(F_n)_{n \in \mathbb{N}_0}$ satisfy $F_{n+2} = F_{n+1} + F_n$ for every $n \in \mathbb{N}_0$, with $F_0 = F_1 = 1$. Put $\varphi_1 = (1 + \sqrt{5})/2$ and $\varphi_2 = (1 - \sqrt{5})/2$.

- a) **FCR** *Fibonacci Characteristic Roots*. Derive the characteristic equation $k^2 - k - 1 = 0$ and its two roots φ_1, φ_2 .
- b) **BIC** *Binet Initial Coefficients*. Determine $A, B \in \mathbb{R}$ in $F_n = A\varphi_1^n + B\varphi_2^n$ and prove

$$F_n = \frac{\varphi_1^{n+1} - \varphi_2^{n+1}}{\sqrt{5}} \quad (n \in \mathbb{N}_0).$$

Use $\varphi_1\varphi_2 = -1$ to recover the alternative form in the original notes.

C. — Damping and sustained forcing

Damped oscillator model. Let $M, k \in (0, \infty)$ be a mass and spring constant, let $l \in [0, \infty)$ be a damping coefficient, let $F: [0, \infty) \rightarrow \mathbb{R}$ be a continuous applied force, and let $x: [0, \infty) \rightarrow \mathbb{R}$ be a twice continuously differentiable vertical displacement. The restoring and damping forces are $-kx$ and $-l\dot{x}$.

57)

- a) **NFM** *Newton Force Model*. Apply Newton's second law to obtain

$$M\ddot{x} + l\dot{x} + kx = F(t).$$

- b) **NDS** *Nondimensional Derivative Scaling*. Define

$$\begin{aligned} \tau &= t\sqrt{k/M}, & \kappa &= \frac{l}{2\sqrt{kM}}, \\ X(\tau) &= x\left(\tau\sqrt{M/k}\right), & f(\tau) &= \frac{F\left(\tau\sqrt{M/k}\right)}{k}. \end{aligned}$$

Transform derivatives carefully and derive $X'' + 2\kappa X' + X = f(\tau)$. For the rest of Part C, relabel X as x and let dots denote differentiation with respect to τ .

- c) **FRC** *Free Response Characteristic equation*. For $f = 0$, derive the roots $-\kappa \pm \sqrt{\kappa^2 - 1}$.

58) Assume $\kappa > 0$.

- a) **UOD** *Underdamped Oscillatory Decay*. If $0 < \kappa < 1$, prove

$$x(\tau) = e^{-\kappa\tau} \left(A \sin(\sqrt{1 - \kappa^2} \tau) + B \cos(\sqrt{1 - \kappa^2} \tau) \right).$$

Here $A, B \in \mathbb{R}$. Identify the oscillation period and envelope decay time.

- b) **CDR** *Critically Damped Response*. If $\kappa = 1$, derive $x(\tau) = (A + B\tau)e^{-\tau}$, where $A, B \in \mathbb{R}$, and identify its dimensionless and dimensional rise and decay scales.
- c) **ODTS** *Overdamped Dual Time Scales*. If $\kappa > 1$, put $\lambda_1 = \kappa - \sqrt{\kappa^2 - 1}$ and $\lambda_2 = \kappa + \sqrt{\kappa^2 - 1}$. Prove $x = Ae^{-\lambda_1\tau} + Be^{-\lambda_2\tau}$, where $A, B \in \mathbb{R}$, and identify the slow decay and fast rise scales.
- d) **TGPD** *Transient Growth Precedes Decay*. Give choices of A, B in either the critical or overdamped case for which $|x|$ initially increases even though $x(\tau) \rightarrow 0$ as $\tau \rightarrow \infty$.

59) Let $\kappa \in (0, \infty)$ and consider $\ddot{x} + 2\kappa\dot{x} + x = \sin \tau$, where dots denote differentiation with respect to τ .

- a) **SFPI** *Sinusoidal Forcing Particular Integral*. Substitute $x_p = C \sin \tau + D \cos \tau$ and prove $x_p = -(2\kappa)^{-1} \cos \tau$.

- b) **TAA** *Transient Approaches Asymptotic response.* Add the appropriate complementary function from Question 58. Prove that it tends to zero and hence

$$x(\tau) = -\frac{1}{2\kappa} \cos \tau + o(1) \quad (\tau \rightarrow \infty).$$

Describe the phase relation between forcing and long-time response.

D. — Impulses and step inputs

Impulse and nascent delta families. The *impulse* of a force F acting near time T is its time integral. A *nascent delta family* is a family $D_\varepsilon: \mathbb{R} \rightarrow [0, \infty)$, indexed by $\varepsilon > 0$, such that $D_\varepsilon(t) \rightarrow 0$ for every $t \neq 0$ and $\int_{-\infty}^{\infty} D_\varepsilon(t) dt \rightarrow 1$ as $\varepsilon \rightarrow 0^+$.

Admitted result. **GIN** *Gaussian Integral Normalization.* The improper integral

$$\int_{-\infty}^{\infty} e^{-u^2} du$$

exists and equals $\sqrt{\pi}$.

60)

- a) **MJI** *Momentum Jump from an Impulse.* Let $T \in \mathbb{R}$ and $\varepsilon > 0$. Suppose a ball of mass $m > 0$ has a twice continuously differentiable position $x: [T - \varepsilon, T + \varepsilon] \rightarrow \mathbb{R}$ and is subject to a continuous contact force $F: [T - \varepsilon, T + \varepsilon] \rightarrow \mathbb{R}$. If $m\ddot{x} = F(t) - mg$ on this interval, where $g > 0$, integrate and prove

$$m\dot{x}(T + \varepsilon) - m\dot{x}(T - \varepsilon) = I_\varepsilon - 2\varepsilon mg,$$

where $I_\varepsilon = \int_{T-\varepsilon}^{T+\varepsilon} F(t) dt$.

- b) **GND** *Gaussian Nascent Delta.* For $\varepsilon > 0$, define

$$D_\varepsilon(t) = \frac{1}{\varepsilon\sqrt{\pi}} e^{-t^2/\varepsilon^2}.$$

Using the substitution $u = t/\varepsilon$ and GIN, prove that its integral is 1, its maximum height is $D_\varepsilon(0) = 1/(\varepsilon\sqrt{\pi})$, and $D_\varepsilon(\pm\varepsilon) = e^{-1}D_\varepsilon(0)$. Deduce that its height is $O(1/\varepsilon)$ and its e-folding width is $O(\varepsilon)$, and prove that it tends pointwise to 0 away from the origin.

Admitted result. **DSP** *Dirac Sifting Property.* The Dirac delta δ is the generalized object defined by its action under integration. If $a, b, c \in \mathbb{R}$ satisfy $a < b$ and $g: [a, b] \rightarrow \mathbb{R}$ is continuous, then

$$\int_a^b g(x)\delta(x-c) dx = \begin{cases} g(c), & c \in (a, b), \\ 0, & c \notin [a, b]. \end{cases}$$

Endpoint conventions require a separately specified regularization. The identity represents the corresponding limit against a standard nascent delta family and is used only inside integrals.

If the impulses I_ε in MJI tend to $I \in \mathbb{R}$ as $\varepsilon \rightarrow 0^+$, the corresponding zero-duration contact model is $m\ddot{x} = -mg + I\delta(t - T)$ in the generalized sense of DSP.

- 61) Consider, in the generalized sense specified by DSP,

$$y'' - y = 3\delta(x - \pi/2), \quad y(0) = y(\pi) = 0,$$

and require y to be continuous on $[0, \pi]$ and twice differentiable away from $x = \pi/2$.

- a) **PRHS** *Piecewise Regional Homogeneous Solutions.* Solve the homogeneous equation on each side and use the endpoint conditions to write

$$y(x) = A \sinh x \quad (0 \leq x < \pi/2), \quad y(x) = C \sinh(\pi - x) \quad (\pi/2 < x \leq \pi).$$

- b) **CCJ** *Continuity Condition and Jump.* Use continuity to prove $A = C$. Integrate the equation across $[\pi/2 - \eta, \pi/2 + \eta]$, let $\eta \rightarrow 0^+$, and prove

$$y'(\pi/2^+) - y'(\pi/2^-) = 3.$$

- c) **BVDC** *Boundary Values Determine the Constants.* Deduce

$$A = C = -\frac{3}{2 \cosh(\pi/2)},$$

and state the resulting piecewise solution, including its common value at $x = \pi/2$.

- d) **RDHS** *Regularity Drop at the Highest-order Source.* Explain why y is continuous while y' jumps, and why allowing y itself to jump would introduce a more singular derivative than the prescribed delta forcing.

Heaviside step. The *Heaviside step function* is

$$H(x) = \begin{cases} 0, & x < 0, \\ 1, & x > 0, \end{cases}$$

with its value at 0 left unspecified. It represents an input switched on at the origin.

62)

- a) **HSD** *Heaviside Step Definition.* Use DSP formally to justify $H(x) = \int_{-\infty}^x \delta(t) dt$ for $x \neq 0$.
- b) **DDH** *Delta Derivative of Heaviside.* Let $a, b \in \mathbb{R}$ satisfy $a < 0 < b$, and let $g: [a, b] \rightarrow \mathbb{R}$ be continuously differentiable with $g(a) = g(b) = 0$. Prove by integration by parts on $[a, 0]$ and $(0, b]$ that

$$-\int_a^b H(x)g'(x) dx = g(0).$$

Compare with DSP and explain the generalized identity $H' = \delta$.

End of Part VI

Part VII. — Series solutions near ordinary and singular points

This part makes the formal power-series method systematic. The convergence and existence theorems stated in the original notes remain admitted results; the coefficient calculations themselves are developed in the questions.

A. — Classification of expansion points

Analytic functions and singular points. Let $I \subseteq \mathbb{R}$ be an open interval and let $x_0 \in I$. A function $h: I \rightarrow \mathbb{R}$ is *analytic at x_0* if there are $\rho > 0$ and coefficients $(a_n)_{n \in \mathbb{N}_0}$ such that $(x_0 - \rho, x_0 + \rho) \subseteq I$ and $h(x) = \sum_{n=0}^{\infty} a_n(x - x_0)^n$ throughout that interval, with the series convergent there. Let $p, q, r: I \rightarrow \mathbb{R}$, suppose that p is non-zero on some punctured neighbourhood of x_0 , and consider

$$p(x)y'' + q(x)y' + r(x)y = 0$$

for an unknown twice differentiable function y . The point x_0 is an *ordinary point* if q/p and r/p , initially defined on the punctured neighbourhood, extend to functions analytic at x_0 . Otherwise it is a *singular point*. A singular point is *regular singular* if $(x - x_0)q(x)/p(x)$ and $(x - x_0)^2r(x)/p(x)$ extend analytically to x_0 ; otherwise it is *irregular singular*. This is equivalent to the $P(x)(x - x_0)^2y'' + Q(x)(x - x_0)y' + R(x)y = 0$ form used in the source.

63)

- a) **OSPD** *Ordinary and Singular Point Definition.* Explain why a zero of p is a candidate singular point but why the analyticity of the two quotients, rather than $p(x_0) \neq 0$ alone, is the definition.
- b) **RSPT** *Regular Singularity through Pole Tests.* Show that a singular point is regular singular precisely when q/p has at most a simple pole and r/p has at most a double pole there, with the pole language interpreted through analytic extension.
- c) **CPE** *Classification of Principal Examples.* Classify all real points for each equation:

$$\begin{aligned}(1 - x^2)y'' - 2xy' + 2y &= 0, \\ (\sin x)y'' + (\cos x)y' + 2y &= 0, \\ (1 + \sqrt{x})y'' - 2xy' + 2y &= 0 \quad (x \geq 0).\end{aligned}$$

In particular, prove that 0 is ordinary and ± 1 are regular singular in the first equation; every $n\pi$, $n \in \mathbb{Z}$, is regular singular in the second; and 0 is irregular singular in the third because the required coefficients are not analytic there.

Admitted result. **OPSB** *Ordinary-Point Series Basis.* Let $I \subseteq \mathbb{R}$ be an open interval, let $x_0 \in I$, and let $P, Q: I \rightarrow \mathbb{R}$ be analytic at x_0 . There is an open interval $J \subseteq I$ containing x_0 on which $y'' + P(x)y' + Q(x)y = 0$ has two linearly independent solutions represented by convergent power series $\sum_{n=0}^{\infty} a_n(x - x_0)^n$. Equivalently, for every $u_0, v_0 \in \mathbb{R}$ there is exactly one such local analytic solution satisfying $y(x_0) = u_0$ and $y'(x_0) = v_0$.

Frobenius series and indicial equation. At a regular singular point x_0 , the *Frobenius method* substitutes

$$y = \sum_{n=0}^{\infty} a_n(x - x_0)^{n+\sigma}, \quad a_0 \neq 0.$$

The lowest-power coefficient gives the *indicial equation* for the *index* σ .

Admitted result. **FRSE** *Frobenius Regular-Singularity Existence.* Let $I \subseteq \mathbb{R}$ be an open interval, let $x_0 \in I$, and let $p, q, r: I \rightarrow \mathbb{R}$ be such that x_0 is a regular singular point

of $p(x)y'' + q(x)y' + r(x)y = 0$. There exist $\rho > 0$, an index $\sigma \in \mathbb{C}$, and coefficients $(a_n)_{n \in \mathbb{N}_0}$ in $\mathbb{C}$, with $a_0 \neq 0$ and $(x_0, x_0 + \rho) \subseteq I$, for which

$$y(x) = (x - x_0)^\sigma \sum_{n=0}^{\infty} a_n (x - x_0)^n,$$

defines a non-zero complex-valued solution for $x_0 < x < x_0 + \rho$. The power series converges for $|x - x_0| < \rho$, and σ is a root of the indicial equation.

B. — Ordinary-point recurrences

64) Seek a power-series solution about 0 of

$$(1 - x^2)y'' - 2xy' + 2y = 0$$

in the form $y = \sum_{n=0}^{\infty} a_n x^n$.

- a) **OPSS** *Ordinary-Point Series Substitution*. Write series for y' and y'' , align their powers of x , and prove that the coefficient of x^n vanishes precisely when

$$(n+2)(n+1)a_{n+2} + (2 - n(n+1))a_n = 0 \quad (n \in \mathbb{N}_0).$$

- b) **LCR** *Legendre Coefficient Recurrence*. Deduce $a_{n+2} = (n-1)a_n/(n+1)$. Treat $n=0$ and $n=1$ before making any division, and explain why a_0 and a_1 are free.
- c) **EOA** *Even and Odd Alternatives*. Prove that the odd branch terminates at a_1x , while

$$a_{2k} = -\frac{a_0}{2k-1} \quad (k \in \mathbb{N}).$$

Hence obtain

$$y = a_0 \left(1 - x^2 - \frac{x^4}{3} - \frac{x^6}{5} - \cdots \right) + a_1 x.$$

- d) **LSF** *Logarithmic Summation Form*. For $|x| < 1$, use

$$\log \left(\frac{1+x}{1-x} \right) = 2 \sum_{k=0}^{\infty} \frac{x^{2k+1}}{2k+1}$$

to sum the even branch and prove

$$y = a_0 \left[1 - \frac{x}{2} \log \left(\frac{1+x}{1-x} \right) \right] + a_1 x.$$

Relate the logarithmic singularities to the regular singular points $x = \pm 1$.

C. — Frobenius indices and resonance

65) Consider

$$4xy'' + 2(1-x^2)y' - xy = 0$$

on $(0, \infty)$ near 0.

- a) **RSC** *Regular Singularity Check*. Multiply by x and show that the normalized coefficients required in the regular-singular definition are $(1-x^2)/2$ and $-x^2/4$.
- b) **GFR** *General Frobenius Recurrence*. Substitute $y = \sum_{n=0}^{\infty} a_n x^{n+\sigma}$, using $a_{-2} = a_{-1} = 0$, and derive

$$2(n+\sigma)(2n+2\sigma-1)a_n = (2n+2\sigma-3)a_{n-2} \quad (n \in \mathbb{N}_0).$$

- c) **IRI** *Indicial Roots Identification*. Set $n=0$ before dividing and prove that $2\sigma(2\sigma-1)a_0 = 0$, so $\sigma = 0$ or $\sigma = 1/2$.

- d) **ASB** *Analytic Series Branch*. For $\sigma = 0$, prove $a_1 = 0$ and

$$a_n = \frac{2n-3}{2n(2n-1)} a_{n-2} \quad (n \geq 2).$$

Calculate the coefficients of x^2 and x^4 and recover the analytic branch printed in the original notes.

- e) **HIB** *Half-Integer Branch*. For $\sigma = 1/2$, prove $a_1 = 0$ and

$$a_n = \frac{n-1}{n(2n+1)} a_{n-2} \quad (n \geq 2).$$

Calculate the first two non-constant coefficients and obtain the branch $x^{1/2}(1 + x^2/10 + x^4/120 + \cdots)$ up to an arbitrary non-zero factor.

Admitted result. **FSRD** *Frobenius Solutions by Root Difference*. Let $I \subseteq \mathbb{R}$ be an open interval, let $x_0 \in I$, and let $p, q, r: I \rightarrow \mathbb{R}$ be such that x_0 is a regular singular point of $p(x)y'' + q(x)y' + r(x)y = 0$. Let $\sigma_1, \sigma_2 \in \mathbb{C}$ be the roots of its indicial equation, ordered so that $\operatorname{Re} \sigma_1 \leq \operatorname{Re} \sigma_2$. If $\sigma_2 - \sigma_1$ is not an integer, two linearly independent Frobenius series solutions arise from the two roots on a sufficiently small interval to the right of x_0 . If the difference is an integer, including zero, a Frobenius series solution y_1 exists for the larger root and a second independent solution generally has the form

$$y_2 = C \log(x - x_0) y_1 + (x - x_0)^{\sigma_1} \sum_{n=0}^{\infty} b_n (x - x_0)^n.$$

Here $C \in \mathbb{C}$, the coefficients $b_n \in \mathbb{C}$, and the displayed series converges on that interval. The constant C can vanish, in which case two log-free Frobenius solutions may exist even though the root difference is an integer.

- 66) Consider $x^2 y'' - xy = 0$ on $(0, \infty)$ near 0.

- a) **RFRE** *Resonant Frobenius Recurrence Equation*. Substitute $y = \sum_{n=0}^{\infty} a_n x^{n+\sigma}$, with $a_{-1} = 0$, and derive

$$(n + \sigma)(n + \sigma - 1) a_n = a_{n-1} \quad (n \in \mathbb{N}_0).$$

Obtain the indicial roots 0 and 1.

- b) **URB** *Upper Root Branch*. For $\sigma = 1$, prove $a_n = a_0 / ((n+1)(n!)^2)$ and hence

$$y_1 = a_0 x \left(1 + \frac{x}{2} + \frac{x^2}{12} + \frac{x^3}{144} + \cdots \right).$$

- c) **LRF** *Lower Root Failure*. For $\sigma = 0$, inspect the recurrence first at $n = 0$ and then at $n = 1$. Prove that it would require $0 \cdot a_1 = a_0$, contradicting $a_0 \neq 0$. Explain why setting $a_0 = 0$ merely reproduces the upper-root solution.
- d) **LSS** *Logarithmic Second Solution*. Apply FSRD and the failure in part (c) to explain why the logarithmic coefficient in a second independent solution is non-zero. After rescaling that solution, state the permitted form

$$y_2 = y_1 \log x + \sum_{n=0}^{\infty} b_n x^n.$$

Explain how substitution would determine the coefficients b_n and why the logarithm represents resonance between the two indices.

End of Part VII

Part VIII. — Gradients, Taylor forms, and stationary points

This part returns to functions of several variables. It extracts directional information from the total differential and uses the second-order term to classify stationary points.

A. — Gradients and directional derivatives

Euclidean vectors and the gradient. For a positive integer $n \in \mathbb{N}$ and vectors $u = (u_1, \dots, u_n), v = (v_1, \dots, v_n) \in \mathbb{R}^n$, define $u \cdot v = \sum_{j=1}^n u_j v_j$ and $|u| = \sqrt{u \cdot u}$. A vector $s \in \mathbb{R}^n$ is a unit vector if $|s| = 1$. Let $U \subseteq \mathbb{R}^n$ be open and let $f: U \rightarrow \mathbb{R}$ have first partial derivatives. Its *gradient* is

$$\nabla f(x) = \left(\frac{\partial f}{\partial x_1}(x), \dots, \frac{\partial f}{\partial x_n}(x) \right) \in \mathbb{R}^n.$$

It is also written $\text{grad } f$. For $x_0 \in U$ and a unit vector $s \in \mathbb{R}^n$, the *directional derivative* is

$$D_s f(x_0) = \lim_{h \rightarrow 0} \frac{f(x_0 + hs) - f(x_0)}{h},$$

provided the limit exists.

Admitted result. CSI Cauchy–Schwarz Inequality. For every $n \in \mathbb{N}$ and all $u, v \in \mathbb{R}^n$,

$$|u \cdot v| \leq |u| |v|,$$

with equality if and only if u and v are linearly dependent.

67) Let $f: U \rightarrow \mathbb{R}$ be differentiable on U , and fix $x_0 \in U$.

- a) **TDGGV** *Total Differential Gives the Gradient Vector.* For an infinitesimal displacement $ds = (dx_1, \dots, dx_n)$, rewrite the total differential as

$$df = ds \cdot \nabla f(x_0).$$

- b) **DDGL** *Directional Derivative as a Gradient Limit.* Substitute the displacement hs in the linear approximation and prove

$$D_s f(x_0) = s \cdot \nabla f(x_0).$$

- c) **MRG** *Maximum Rate from the Gradient.* Use the Cauchy–Schwarz inequality to prove $|D_s f(x_0)| \leq |\nabla f(x_0)|$ for every unit vector s . If the gradient is non-zero, determine the directions of maximum increase and maximum decrease.
- d) **GOLC** *Gradient Orthogonality to Level Curves.* Let $I \subseteq \mathbb{R}$ be an open interval, let $\gamma: I \rightarrow U$ be differentiable, and suppose $f(\gamma(t)) = c$ for every $t \in I$, where $c \in \mathbb{R}$. Differentiate along the path and prove $\gamma'(t) \cdot \nabla f(\gamma(t)) = 0$. Interpret this as orthogonality between the gradient and every regular tangent to a contour.

Stationary points and saddles. A point $x_0 \in U$ is a *stationary point*, or *critical point*, if $\nabla f(x_0) = 0$. It is a *local minimum* if there is a radius $\rho > 0$ such that $f(x) \geq f(x_0)$ whenever $x \in U$ and $|x - x_0| < \rho$, and a *strict local minimum* if the inequality is strict for $x \neq x_0$. It is a *local maximum* if the same condition holds with $f(x) \leq f(x_0)$, with the analogous *strict* variant, and a *saddle point* if it is stationary but neither a local minimum nor a local maximum.

68)

- a) **AVSD** *All Vanishing Stationary Directions.* Let $f: U \rightarrow \mathbb{R}$ be differentiable at a stationary point $x_0 \in U$. Prove that every directional derivative of f at x_0 vanishes.
- b) **SDD** *Saddle Direction Demonstration.* For $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ defined by $f(x, y) = x^2 - y^2$, prove that $(0, 0)$ is stationary and exhibit directions along which f increases and decreases. Describe the two crossing lines of the zero contour.

B. — The Hessian and multivariable Taylor expansion

Hessian matrix. Let $U \subseteq \mathbb{R}^n$ be open and let $f: U \rightarrow \mathbb{R}$ have continuous second partial derivatives. Its *Hessian matrix* at $x \in U$ is

$$H_f(x) = \left(\frac{\partial^2 f}{\partial x_i \partial x_j}(x) \right)_{1 \leq i, j \leq n}.$$

By SCECP it is a real symmetric matrix. A stationary point x_0 is *non-degenerate* if $H_f(x_0)$ is invertible, that is, if zero is not an eigenvalue of $H_f(x_0)$.

69) Let $x_0 \in U$ and $h \in \mathbb{R}^n$ be sufficiently small that $x_0 + th \in U$ for every $t \in [0, 1]$. Define $g: [0, 1] \rightarrow \mathbb{R}$ by $g(t) = f(x_0 + th)$.

- a) **LPR** *Line Parameter Restriction.* Apply the multivariable chain rule to prove $g'(t) = h \cdot \nabla f(x_0 + th)$.
- b) **LRSD** *Line-Restricted Second Derivative.* Differentiate once more and prove

$$g''(t) = h^\top H_f(x_0 + th)h = \sum_{i=1}^n \sum_{j=1}^n h_i h_j \frac{\partial^2 f}{\partial x_i \partial x_j}(x_0 + th).$$

- c) **SOTQ** *Second-Order Term as a Quadratic form.* In dimension two, with $h = (\delta x, \delta y)$, expand $h^\top H_f(x_0)h$ as

$$f_{xx}(\delta x)^2 + 2f_{xy}\delta x\delta y + f_{yy}(\delta y)^2,$$

where all derivatives are evaluated at x_0 .

- d) **MTE** *Multivariable Taylor Expansion.* Apply the Fundamental Theorem of Calculus twice to prove

$$g(1) = g(0) + g'(0) + \int_0^1 (1-t)g''(t) dt.$$

Use the continuity of the Hessian at x_0 to deduce

$$f(x_0 + h) = f(x_0) + h \cdot \nabla f(x_0) + \frac{1}{2}h^\top H_f(x_0)h + o(|h|^2)$$

as $h \rightarrow 0$.

C. — Classification and contours

Definiteness. For a real symmetric $n \times n$ matrix H , its quadratic form is $Q_H(h) = h^\top Hh$. It is *positive definite* if $Q_H(h) > 0$ for every $h \neq 0$, *negative definite* if $Q_H(h) < 0$ for every $h \neq 0$, and *indefinite* if it takes both positive and negative values.

Admitted result. **SSTD** *Symmetric Spectral Theorem and Diagonalization.* Let $n \in \mathbb{N}$. For every real symmetric $n \times n$ matrix H , there is an orthonormal basis of $\mathbb{R}^n$ consisting of eigenvectors of H . In the corresponding coordinates, H is diagonal with real eigenvalues $\lambda_1, \dots, \lambda_n$, and

$$h^\top Hh = \sum_{j=1}^n \lambda_j X_j^2,$$

where $X_1, \dots, X_n$ are the coordinates of h in that basis.

70) Let x_0 be a stationary point of a function $f: U \rightarrow \mathbb{R}$ having the Taylor expansion from Question 69.

- a) **QFDC** *Quadratic Form Definiteness Cases.* Use SSTD to prove that a positive-definite Hessian satisfies $h^\top H_f(x_0)h \geq m|h|^2$ for some $m > 0$ and every $h \in \mathbb{R}^n$. Use the remainder definition to deduce a strict local minimum. Give the corresponding argument for a negative-definite Hessian and a strict local maximum.
- b) **SEM** *Spectral Eigenvalue Method.* Apply SSTD to prove that the Hessian is positive definite exactly when all its eigenvalues are positive, negative definite exactly when they are all negative, and indefinite when it has eigenvalues of both signs.
- c) **ISPC** *Indefinite Sign Produces a Crossing.* If the Hessian is indefinite, use two eigenvector directions and the Taylor expansion to prove that x_0 is a saddle point.
- d) **ZEC** *Zero Eigenvalue Caveat.* Explain why a zero eigenvalue makes the quadratic test inconclusive. Illustrate this with the one-variable functions $x \mapsto x^4$, $x \mapsto -x^4$, and $x \mapsto x^3$ at 0.

Leading principal minors. Let $n \in \mathbb{N}$. For a real symmetric $n \times n$ matrix H , the k th leading principal minor Δ_k is the determinant of its upper-left $k \times k$ submatrix, for $1 \leq k \leq n$.

Admitted result. SCFD *Sylvester Conditions For Definiteness.* Let $n \in \mathbb{N}$ and let H be a real symmetric $n \times n$ matrix with leading principal minors $\Delta_1, \dots, \Delta_n$. The matrix is positive definite if and only if $\Delta_k > 0$ for every $1 \leq k \leq n$. It is negative definite if and only if $(-1)^k \Delta_k > 0$ for every $1 \leq k \leq n$. Failure of both tests need not imply indefiniteness when zero eigenvalues are present.

71)

- a) **LPMD** *Leading Principal Minor Determinants.* For $H = \begin{pmatrix} a & b \\ b & c \end{pmatrix}$, identify $\Delta_1 = a$ and $\Delta_2 = ac - b^2$.
- b) **SCPA** *Sylvester Criterion Positive Application.* Use SCFD to prove that H is positive definite precisely when $a > 0$ and $ac - b^2 > 0$.
- c) **SNA** *Sylvester Negative Application.* Prove the corresponding negative-definite conditions $a < 0$ and $ac - b^2 > 0$.
- d) **ITD** *Indefiniteness in Two Dimensions.* Prove that $ac - b^2 < 0$ makes H indefinite. Explain why equality to zero requires higher-order information in a stationary-point problem.

72) Define $f: \mathbb{R}^2 \rightarrow \mathbb{R}$ by

$$f(x, y) = 4x^3 - 12xy + y^2 + 10y + 6.$$

- a) **SPS** *Solving Polynomial Stationarity.* Compute the gradient and prove that the stationary points are exactly $(1, 1)$ and $(5, 25)$.
- b) **HCPE** *Hessian Computation at Principal Examples.* Prove

$$H_f(x, y) = \begin{pmatrix} 24x & -12 \\ -12 & 2 \end{pmatrix}.$$

At $(1, 1)$ compute a negative determinant and classify a saddle. At $(5, 25)$ compute leading principal minors 120 and 96 and classify a strict local minimum.

- c) **QCC** *Quadratic Contour Classification.* In coordinates aligned with the Hessian eigenvectors, show that the quadratic-level curves near a non-degenerate minimum or maximum are ellipses, while those near a saddle are hyperbolas. Explain the crossing of the saddle's quadratic zero contour at the stationary point.

End of Part VIII

Part IX. — Differential systems and phase portraits

This part packages interacting first-order equations in vector form. It connects eigenvalues with phase portraits and then applies the linear theory locally to nonlinear population dynamics.

A. — Linear systems and eigenmodes

Matrix form. For a positive integer $n \in \mathbb{N}$, let $M = (m_{ij})_{1 \leq i, j \leq n}$ be a real $n \times n$ matrix and let $Y = (y_1, \dots, y_n)^\top \in \mathbb{R}^n$. Define $(MY)_i = \sum_{j=1}^n m_{ij}y_j$. The trace is $\text{tr } M = \sum_{i=1}^n m_{ii}$. A vector $v \in \mathbb{C}^n \setminus \{0\}$ is an eigenvector with eigenvalue $\lambda \in \mathbb{C}$ if $Mv = \lambda v$; the eigenvalues satisfy $\det(M - \lambda I_n) = 0$, where I_n is the $n \times n$ identity matrix.

For the two-dimensional system, let $I \subseteq \mathbb{R}$ be an open interval, let $a, b, c, d \in \mathbb{R}$, let $f_1, f_2: I \rightarrow \mathbb{R}$ be continuous, and let the unknown functions $y_1, y_2: I \rightarrow \mathbb{R}$ be differentiable. Write

$$\dot{Y} = MY + F(t), \quad Y(t) = \begin{pmatrix} y_1(t) \\ y_2(t) \end{pmatrix}, \quad M = \begin{pmatrix} a & b \\ c & d \end{pmatrix}, \quad F(t) = \begin{pmatrix} f_1(t) \\ f_2(t) \end{pmatrix}.$$

The dot denotes differentiation with respect to $t \in I$.

73)

- a) **ESOE** *Elimination gives a Second-Order Equation.* Assume f_1 is differentiable. Differentiate $\dot{y}_1 = ay_1 + by_2 + f_1$, substitute both system equations, and prove

$$\ddot{y}_1 - (a + d)\dot{y}_1 + (ad - bc)y_1 = bf_2 - df_1 + \dot{f}_1.$$

Do not divide by b , so the identity remains valid when $b = 0$.

- b) **SOFS** *Second Order as a First-order System.* Let $\alpha, \beta \in \mathbb{R}$, let $f: I \rightarrow \mathbb{R}$, and let $y: I \rightarrow \mathbb{R}$ be twice differentiable. By setting $y_1 = y$ and $y_2 = \dot{y}$, transform $\ddot{y} + \alpha\dot{y} + \beta y = f$ into

$$\dot{Y} = \begin{pmatrix} 0 & 1 \\ -\beta & -\alpha \end{pmatrix} Y + \begin{pmatrix} 0 \\ f \end{pmatrix}.$$

- c) **SMN** *System Matrix Notation.* Expand the vector equation component by component and verify that it is exactly equivalent to the original pair of scalar equations.

Admitted result. **LSIVPEU** *Linear System Initial-Value Problem Existence and Uniqueness.* Let $n \in \mathbb{N}$, let $I \subseteq \mathbb{R}$ be an open interval, let $A: I \rightarrow \mathbb{R}^{n \times n}$ and $F: I \rightarrow \mathbb{R}^n$ be continuous, let $t_0 \in I$, and let $Y_0 \in \mathbb{R}^n$. There exists a unique continuously differentiable function $Y: I \rightarrow \mathbb{R}^n$ satisfying $\dot{Y}(t) = A(t)Y(t) + F(t)$ for every $t \in I$ and $Y(t_0) = Y_0$.

74) Let $n \in \mathbb{N}$ and let M be a constant real $n \times n$ matrix. Seek complex-valued solutions on $\mathbb{R}$.

- a) **EAS** *Eigenvector Ansatz for a System.* Let $\lambda \in \mathbb{C}$ and $v \in \mathbb{C}^n \setminus \{0\}$. Substitute $Y(t) = ve^{\lambda t}$ in $\dot{Y} = MY$ and prove that it is a non-zero solution precisely when $Mv = \lambda v$.
- b) **DES** *Distinct Eigenvalue Solutions.* Suppose M has n distinct complex eigenvalues $\lambda_1, \dots, \lambda_n$ with eigenvectors $v_1, \dots, v_n$. Prove, by induction on the number of vectors and an application of $M - \lambda_n I_n$, that these eigenvectors are linearly independent. Show that, for $A_1, \dots, A_n \in \mathbb{C}$,

$$Y(t) = \sum_{j=1}^n A_j v_j e^{\lambda_j t}$$

realizes arbitrary initial data in $\mathbb{C}^n$. Apply LSIVPEU to the real and imaginary parts to conclude that this is the general complex solution.

- c) **RCMC** *Real Combinations from Matched Conjugates*. When M is real and a non-real eigenpair is (λ, v) , prove that $(\bar{\lambda}, \bar{v})$ is also an eigenpair and explain how real and imaginary parts of $ve^{\lambda t}$ give real solutions.

75) Let

$$M = \begin{pmatrix} -4 & 24 \\ 1 & -2 \end{pmatrix}, \quad g = \begin{pmatrix} 4 \\ 1 \end{pmatrix},$$

and consider $\dot{Y} = MY + ge^t$ on $\mathbb{R}$.

- a) **CEC** *Coupled Eigenvalue Calculation*. Compute $\det(M - \lambda I)$ and obtain the eigenvalues 2 and -8 .
- b) **CED** *Coupled Eigenvector Directions*. Find corresponding eigenvectors $v_1 = (4, 1)^\top$ and $v_2 = (-6, 1)^\top$.
- c) **EFPI** *Exponential Forcing Particular Integral*. Seek $Y_p = ue^t$, solve $(I_2 - M)u = g$, and prove $u = (-4, -1)^\top$.
- d) **CGS** *Complete General Solution*. Deduce

$$Y(t) = Av_1e^{2t} + Bv_2e^{-8t} - \begin{pmatrix} 4 \\ 1 \end{pmatrix} e^t, \quad A, B \in \mathbb{R}.$$

- e) **SSP** *Saddle State Portrait*. For the homogeneous system, explain why trajectories on the v_2 line approach the origin, trajectories on the v_1 line recede, and general trajectories have the saddle geometry described by these eigendirections.

Admitted result. **MDFE** *Matrix Determinant Flow Equation*. Let $n \in \mathbb{N}$, let $I \subseteq \mathbb{R}$ be an open interval, let $A: I \rightarrow \mathbb{R}^{n \times n}$ be continuous, and let $\Phi: I \rightarrow \mathbb{R}^{n \times n}$ be differentiable and satisfy $\Phi' = A\Phi$. Then $W = \det \Phi$ satisfies $W' = (\operatorname{tr} A)W$. Consequently, for $t, t_0 \in I$,

$$W(t) = W(t_0) \exp \left(\int_{t_0}^t \operatorname{tr} A(s) \, ds \right).$$

In particular, W is either identically zero or nowhere zero.

76)

- a) **TAFC** *Trace Abel Formula Connection*. Let $I \subseteq \mathbb{R}$ be an open interval and let $p, q: I \rightarrow \mathbb{R}$ be continuous. Apply MDFE to the first-order form of $y'' + p(x)y' + q(x)y = 0$ and recover the Abel formula from Question 49.
- b) **IBFS** *Independence of a Basis of Fundamental Solutions*. Let $n \in \mathbb{N}$, let $I \subseteq \mathbb{R}$ be an open interval, let $A: I \rightarrow \mathbb{R}^{n \times n}$ be continuous, and let the columns of Φ be n solutions of $Y' = A(t)Y$. Prove that if they form a basis at one time $t_0 \in I$, they form a basis at every $t \in I$.

B. — Classification of linear phase portraits

77) Let M be a real 2×2 matrix with two distinct eigenvalues, and consider $\dot{Y} = MY$.

- a) **ROSS** *Real Opposite Signs give a Saddle*. Suppose the eigenvalues are real and satisfy $\lambda_1 > 0 > \lambda_2$. Use the eigenmode solution to prove that the origin is a saddle, attracting along the λ_2 eigendirection and repelling along the λ_1 eigendirection.
- b) **RSSN** *Real Same Signs give a Node*. Suppose the eigenvalues are real and have the same sign. Show that the origin is a stable node when both are negative and an unstable node when both are positive. Determine which eigendirection the trajectories become tangent to far from the origin.

- c) **CCSP** *Complex Conjugates give a Spiral Portrait.* Suppose the eigenvalues are $\alpha \pm i\beta$ with $\beta \neq 0$. Use Question 74 to show that real trajectories combine a rotation of angular frequency $|\beta|$ with the factor $e^{\alpha t}$. Classify inward and outward spirals by the sign of α .
- d) **ZRCC** *Zero Real Component gives a Centre.* If $\alpha = 0$ and the matrix is diagonalizable over $\mathbb{C}$, show that non-zero trajectories are closed curves surrounding the origin; after a real linear coordinate change they are ellipses. This case is called a centre.
- e) **SOC** *Spiral Orientation Check.* Explain how the direction of MY at one point on the positive horizontal axis determines whether a spiral is clockwise or counter-clockwise.

C. — Nonlinear autonomous systems

Equilibria and the Jacobian. Let $D \subseteq \mathbb{R}^2$ be open and let $f, g: D \rightarrow \mathbb{R}$ have continuous first partial derivatives. The *autonomous system*

$$\dot{x} = f(x, y), \quad \dot{y} = g(x, y)$$

has an *equilibrium*, or *fixed point*, at $(x_0, y_0) \in D$ when $f(x_0, y_0) = g(x_0, y_0) = 0$. Its *Jacobian matrix* there is

$$J(x_0, y_0) = \begin{pmatrix} f_x & f_y \\ g_x & g_y \end{pmatrix}_{(x_0, y_0)}.$$

78) Let (x_0, y_0) be an equilibrium and put $x = x_0 + \xi$, $y = y_0 + \eta$.

- a) **PTE** *Perturbation Taylor Expansion.* Apply the first-order multivariable expansion to prove

$$\begin{pmatrix} \dot{\xi} \\ \dot{\eta} \end{pmatrix} = J(x_0, y_0) \begin{pmatrix} \xi \\ \eta \end{pmatrix} + R(\xi, \eta),$$

where R is defined near $(0, 0)$ with values in $\mathbb{R}^2$ and

$$\frac{|R(\xi, \eta)|}{\sqrt{\xi^2 + \eta^2}} \longrightarrow 0 \quad \text{as } (\xi, \eta) \rightarrow (0, 0).$$

- b) **JLM** *Jacobian Linear Model.* Define the *linearised system* by discarding the smaller-order remainder. Explain why its eigenvalues are the candidates controlling local stability.

Admitted result. **HLP** *Hyperbolic Linearization Principle.* Let $n \in \mathbb{N}$, let $D \subseteq \mathbb{R}^n$ be open, let $F: D \rightarrow \mathbb{R}^n$ be continuously differentiable, and let $x_0 \in D$ satisfy $F(x_0) = 0$. If no eigenvalue of the Jacobian $DF(x_0)$ has zero real part, then in a sufficiently small neighbourhood the nonlinear system $\dot{x} = F(x)$ has the same qualitative equilibrium type and stability as its linearisation: if every eigenvalue has negative real part the equilibrium is attracting, if every eigenvalue has positive real part it is repelling, and saddles retain stable and unstable directions. If an eigenvalue has zero real part, this conclusion is not guaranteed and nonlinear terms must be examined.

79) Let $I \subseteq \mathbb{R}$ be an interval and let the differentiable functions $x, y: I \rightarrow [0, \infty)$ denote prey and predator populations. For parameters $\alpha, \beta, \gamma, \varepsilon, \delta > 0$, the model

$$\dot{x} = \alpha x - \beta x^2 - \gamma xy, \quad \dot{y} = \varepsilon xy - \delta y$$

combines prey growth, prey competition, predation, predator reproduction, and predator death. Consider the particular system

$$\dot{x} = 8x - 2x^2 - 2xy, \quad \dot{y} = xy - y.$$

- a) **PPM** *Predator-Prey Modelling terms.* Identify the sign and biological role of every term in the general model.
- b) **EEP** *Ecological Equilibrium Points.* Factor the two right-hand sides of the particular system and prove that its equilibria in $[0, \infty)^2$ are exactly $(0, 0)$, $(4, 0)$, and $(1, 3)$.
- c) **OJS** *Origin Jacobian Saddle.* Compute $J(0, 0)$, find its eigenvalues 8 and -1 , and classify the origin using HLP.
- d) **BJS** *Boundary Jacobian Saddle.* Compute $J(4, 0) = \begin{pmatrix} -8 & -8 \\ 0 & 3 \end{pmatrix}$, find eigenvectors $(1, 0)^\top$ and $(8, -11)^\top$, and classify the point.
- e) **IJS** *Interior Jacobian Spiral.* Compute $J(1, 3) = \begin{pmatrix} -2 & -2 \\ 3 & 0 \end{pmatrix}$ and prove that its eigenvalues are $-1 \pm i\sqrt{5}$. Deduce that $(1, 3)$ is a stable spiral.
- f) **SOD** *Spiral Orientation Direction.* At a perturbation $(\xi, \eta) = (\xi, 0)$ with $\xi > 0$, compute the linearised velocity $(-2\xi, 3\xi)$. Deduce that the stable spiral is counter-clockwise.

End of Part IX

Part X. — Characteristics, waves, and diffusion

This final part extends the evolution viewpoint to functions of two independent variables. It first follows information along characteristic curves, then separates left- and right-travelling waves, and finally derives a similarity profile for heat diffusion.

Partial differential equations. A *partial differential equation*, or *PDE*, is an equation involving partial derivatives of an unknown function of at least two independent variables. In this part, $x \in \mathbb{R}$ denotes position and $t \in [0, \infty)$ denotes time. The first-order wave equation is also called the advection equation; the second-order wave equation is the prototype of a hyperbolic PDE; and the diffusion equation is the prototype of a parabolic PDE. Another standard example is Laplace's equation, which models electrostatic and potential-flow fields.

A. — First-order waves and characteristics

First-order wave equation. Let $c \in \mathbb{R}$ and let $y: \mathbb{R} \times [0, \infty) \rightarrow \mathbb{R}$ have continuous first partial derivatives. The *first-order wave equation* in the sign convention of the original notes is

$$y_t - cy_x = 0.$$

A *characteristic* is a curve in the (x, t) -plane along which the PDE reduces to an ordinary differential equation.

80)

- a) **PCD** *Path Chain-rule Derivative.* Let $x: [0, \infty) \rightarrow \mathbb{R}$ be differentiable. Apply the chain rule to prove

$$\frac{d}{dt}y(x(t), t) = y_x(x(t), t)x'(t) + y_t(x(t), t).$$

- b) **CLCC** *Characteristic Lines Carry Constants.* Use the PDE and choose $x'(t) = -c$. Prove that $x(t) + ct$ and $y(x(t), t)$ are both constant along the resulting straight characteristics.
- c) **GSA** *General Solution of Advection.* Deduce that every continuously differentiable solution has the form

$$y(x, t) = f(x + ct)$$

for a differentiable function $f: \mathbb{R} \rightarrow \mathbb{R}$. Prove the converse by direct differentiation and interpret the sign and speed of translation.

81) In the following items, seek a continuously differentiable function $y: \mathbb{R} \times [0, \infty) \rightarrow \mathbb{R}$.

- a) **TIPA** *Translated Initial Profile under Advection.* Impose $y(x, 0) = x^2 - 3$ for every $x \in \mathbb{R}$ in the preceding equation and obtain $y(x, t) = (x + ct)^2 - 3$.
- b) **FCO** *Forced Characteristic Ordinary equation.* For

$$y_t + 5y_x = e^{-t}, \quad y(x, 0) = e^{-x^2},$$

derive the characteristics $x(t) = x_0 + 5t$ and the ordinary equation $dy/dt = e^{-t}$ along each one.

- c) **FIVP** *Forced Initial-Value Profile.* Integrate the characteristic equation, determine its constant from the initial profile, and prove

$$y(x, t) = 1 + e^{-(x-5t)^2} - e^{-t}.$$

Verify the PDE and initial condition directly.

B. — The second-order wave equation

Wave equation and its string model. Let $c \in (0, \infty)$ and let $y: \mathbb{R} \times [0, \infty) \rightarrow \mathbb{R}$ have continuous second partial derivatives. The *wave equation* is

$$y_{tt} = c^2 y_{xx}.$$

For a vibrating string, $y(x, t)$ is transverse displacement and a positive mass density $\rho: \mathbb{R} \rightarrow (0, \infty)$ converts acceleration into force per unit length. In the uniform model, take $\rho(x) = \rho_0 > 0$ and the tension coefficient $K > 0$ to be constant.

82)

- a) **STI** *String Translation Invariance.* Explain why a restoring law for a uniform string cannot depend on the absolute value of y : translating the whole string vertically cannot create a force.
- b) **USBF** *Uniform Slope gives Balanced Forces.* Explain why a constant slope y_x also gives no net transverse force on an interior string element.
- c) **RFC** *Restoring Force from Curvature.* Deduce why the first local spatial derivative capable of producing a net restoring force is the curvature y_{xx} . Using the model $\rho_0 y_{tt} = K y_{xx}$, recover the wave equation and identify $c^2 = K/\rho_0$ as the force-to-density ratio.

83)

- a) **OFD** *Operator Factorization by Difference.* Prove the commuting-operator identity

$$\partial_t^2 - c^2 \partial_x^2 = (\partial_t + c \partial_x)(\partial_t - c \partial_x).$$

- b) **WNC** *Wave Nullspace Components.* Verify that $f(x + ct)$ is annihilated by $\partial_t - c \partial_x$ and $g(x - ct)$ by $\partial_t + c \partial_x$, for twice differentiable $f, g: \mathbb{R} \rightarrow \mathbb{R}$.
- c) **CCT** *Characteristic Coordinate Transformation.* Define $\xi = x + ct$ and $\eta = x - ct$. Use the chain rule to prove

$$y_{tt} - c^2 y_{xx} = -4c^2 y_{\xi\eta}.$$

Verify that the coordinate change is invertible by expressing x and t in terms of ξ and η .

- d) **WDAS** *Wave Decomposition into Advancing Signals.* Integrate $y_{\xi\eta} = 0$ first with respect to one variable and then the other to prove the general form

$$y(x, t) = f(x + ct) + g(x - ct).$$

Identify the left- and right-travelling components.

Data for a wave equation. On the whole real line, prescribe an initial displacement $\phi: \mathbb{R} \rightarrow \mathbb{R}$ and initial velocity $\psi: \mathbb{R} \rightarrow \mathbb{R}$. On a bounded spatial interval, the second spatial derivative additionally calls for two boundary conditions, one at each end, while the second time derivative calls for the two initial functions.

84) Assume ϕ is twice continuously differentiable and ψ is continuously differentiable.

- a) **IDR** *Initial Data Relations.* Substitute $t = 0$ in the two-wave representation and prove $f + g = \phi$ and $c(f' - g') = \psi$.
- b) **DIF** *Dual-wave Initial Formula.* Solve these relations, allowing an irrelevant opposite additive constant in f and g , and derive

$$y(x, t) = \frac{\phi(x + ct) + \phi(x - ct)}{2} + \frac{1}{2c} \int_{x-ct}^{x+ct} \psi(s) \, ds.$$

- c) **SVZ** *Symmetric Zero-Velocity waves.* When ψ is the zero function, explain why the initial profile splits into two equal waves travelling in opposite directions.
- d) **TLP** *Travelling Lorentzian Pulses.* Apply part (c) to $\phi(x) = 1/(1+x^2)$ and prove

$$y(x, t) = \frac{1}{2} \left[\frac{1}{1+(x+ct)^2} + \frac{1}{1+(x-ct)^2} \right].$$

Check that $y(x, t) \rightarrow 0$ as $x \rightarrow \pm\infty$ for each fixed $t \geq 0$.

C. — Diffusion and similarity

Diffusion equation and error functions. Let $\kappa \in (0, \infty)$ be the thermal diffusivity. For a temperature $T: [0, \infty) \times (0, \infty) \rightarrow \mathbb{R}$ having continuous derivatives T_t and T_{xx} , the *diffusion equation*, or *heat equation*, is

$$T_t = \kappa T_{xx}.$$

Unlike the wave equation, it relates rate of change directly to curvature, so temperature irregularities smooth rather than propagate unchanged. For $\eta \in \mathbb{R}$, define

$$\operatorname{erf}(\eta) = \frac{2}{\sqrt{\pi}} \int_0^\eta e^{-u^2} du, \quad \operatorname{erfc}(\eta) = 1 - \operatorname{erf}(\eta).$$

The Gaussian integral used in Question 60 gives $\operatorname{erf}(\eta) \rightarrow 1$ as $\eta \rightarrow \infty$. The conditions below model a bar initially at temperature zero whose end is switched to unit temperature at $t = 0$, equivalently by the Heaviside input from Question 62 when negative times are included.

- 85)** Consider a semi-infinite bar satisfying, for $x > 0$ and $t > 0$,

$$\begin{aligned} T_t &= \kappa T_{xx}, & T(0, t) &= 1, \\ T(x, t) &\longrightarrow 0 & \text{as } t \rightarrow 0^+ & \text{for each fixed } x > 0, \\ T(x, t) &\longrightarrow 0 & \text{as } x \rightarrow \infty & \text{for each fixed } t > 0. \end{aligned}$$

- a) **DLS** *Diffusive Length Scale.* Use dimensions $[\kappa] = L^2 T^{-1}$ to show that $\sqrt{\kappa t}$ is the natural penetration length. Define

$$\eta = \frac{x}{2\sqrt{\kappa t}}, \quad T(x, t) = \theta(\eta).$$

- b) **SVD** *Similarity Variable Derivatives.* Compute η_x and η_t , then prove

$$T_{xx} = \frac{1}{4\kappa t} \theta''(\eta), \quad T_t = -\frac{\eta}{2t} \theta'(\eta).$$

- c) **RSO** *Reduced Similarity Ordinary equation.* Substitute in the heat equation and obtain $\theta'' + 2\eta\theta' = 0$.
- d) **EFS** *Error Function Solution.* Treat θ' as the unknown in a first-order equation. Use the integrating factor e^{η^2} and prove $\theta(\eta) = A \operatorname{erf}(\eta) + B$ for constants $A, B \in \mathbb{R}$.
- e) **BCSS** *Boundary Conditions Select the Similarity solution.* Translate the boundary condition at $x = 0$, the far-field condition, and the one-sided initial condition for fixed $x > 0$ into conditions at $\eta = 0$ and $\eta = \infty$. Deduce

$$T(x, t) = \operatorname{erfc}\left(\frac{x}{2\sqrt{\kappa t}}\right).$$

- f) **PDS** *Penetration Distance Scaling.* Fix a temperature level $\Theta \in (0, 1)$. Show that the equation $T(x, t) = \Theta$ determines a constant similarity coordinate and hence a penetration distance $x = O(\sqrt{\kappa t})$. If a bar has length $L > 0$, derive the semi-infinite validity condition $t \ll L^2/\kappa$.

End of Part X and of the problem